



Modelling Financial Markets

Chapter 7: Value-at-Risk and Expected Shortfall



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ **AI:** Artificial Intelligence
- ▣ **AR:** AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARMA:** AutoRegressive Moving Average (model)
- ▣ **BET:** Bucharest Exchange Trading (index)
- ▣ **BIS:** Bank for International Settlements
- ▣ **BNR:** Banca Națională a României — *the National Bank of Romania*
- ▣ **BVB:** Bursa de Valori București — *the Bucharest Stock Exchange*
- ▣ **CF:** Cornish–Fisher (expansion)
- ▣ **COVID-19:** COronaVirus Disease 2019
- ▣ **CQFE:** Conditional Quantile Forecast Encompassing (test)
- ▣ **CV:** Cross-Validation
- ▣ **DM:** Diebold–Mariano (test)
- ▣ **ES:** Expected Shortfall
- ▣ **ETF:** Exchange-Traded Fund
- ▣ **EU:** European Union
- ▣ **EVT:** Extreme Value Theory
- ▣ **FHS:** Filtered Historical Simulation

Acronyms used in this material (2/2)

- ▣ **FRTB**: Fundamental Review of the Trading Book
- ▣ **FX**: Foreign eXchange
- ▣ **FZ0**: Fissler–Ziegel loss, zero-homogeneous version
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GAS**: Generalised Autoregressive Score (model)
- ▣ **GEV**: Generalised Extreme Value (distribution)
- ▣ **GPD**: Generalised Pareto Distribution
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **LASSO**: Least Absolute Shrinkage and Selection Operator
- ▣ **MIDAS**: MlXed DAta Sampling (regression)
- ▣ **MSE**: Mean Squared Error
- ▣ **QML**: Quasi-Maximum Likelihood
- ▣ **RMSFE**: Root Mean Squared Forecast Error
- ▣ **SAV**: Symmetric Absolute Value (CAViaR specification)
- ▣ **SE**: Standard Error
- ▣ **SHAP**: SHapley Additive exPlanations
- ▣ **US**: United States
- ▣ **XFG**: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

Chapter Route and Outcomes

Route of this chapter

- ▣ Part I: definitions, coherence, unconditional estimation methods and their precision
- ▣ Part II: conditional forecasts (FHS, CAViaR), extreme value theory, horizon, portfolio, regulation, a case study and one pass of the AI-assisted research loop
- ▣ Running example: S&P 500 daily losses, 2000–2026; comparisons on BET, Bitcoin, EUR/RON (BNR reference rate) and gold; data up to 18 September 2026 (full inventory in the [Appendix](#))
- ▣ Units: daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$, with P_t the closing price on day t ; technical detail sits in the [Appendix](#)

After this chapter you can

- ▣ define VaR and ES and explain why ES is coherent and VaR is not
- ▣ estimate both historically, parametrically, by filtered simulation and by extreme value theory, each with a standard error or an interval
- ▣ scale them to 10 days, split them across assets and read the Basel and FRTB rules

How Much Can We Lose? RiskMetrics and VaR



J.P. Morgan & Co. building, 23 Wall Street, New York

Photo: [Beyond My Ken \(2012\)](#); CC BY-SA 4.0; [Wikimedia](#)

[Commons](#)

- Tomorrow's portfolio value is random; management and regulators need **one number**: a loss exceeded only rarely, to set limits and **capital**
- J.P. Morgan published **RiskMetrics** in October 1994, a free methodology and data set for market risk; its Technical Document (4th ed., 1996) became the reference for VaR [J.P. Morgan/Reuters \(1996\)](#)
- Textbooks: [Franke, Härdle & Hafner \(2019\)](#), [McNeil, Frey & Embrechts \(2015\)](#), [Danielsson \(2011\)](#)

Losses, Horizons and Tail Probabilities

- Return over one day: $X_{t+1} = r_{t+1}$, the log return in %; **loss** $L_{t+1} = -X_{t+1}$, a **log loss**
 - ▶ Positive L = the position lost money; negative L = a gain
 - ▶ In money, with V_t the value of the position: $\text{loss} = V_t (1 - e^{-L_{t+1}/100}) \approx V_t L_{t+1}/100$ for small L ; with simple returns R , $V_t (-R_{t+1})/100$ is exact
 - ▶ VaR converts exactly through this monotone map; ES must be converted scenario by scenario, then averaged
- **Horizon** h : the number of days over which the loss is measured (1 day, 10 days)
- **Level** α : the **tail probability**, typically $\alpha = 1\%$ or 2.5% : “VaR 1%”, “ES 2.5%”
- **Sign convention in this course**: VaR and ES are **positive** numbers when they are losses

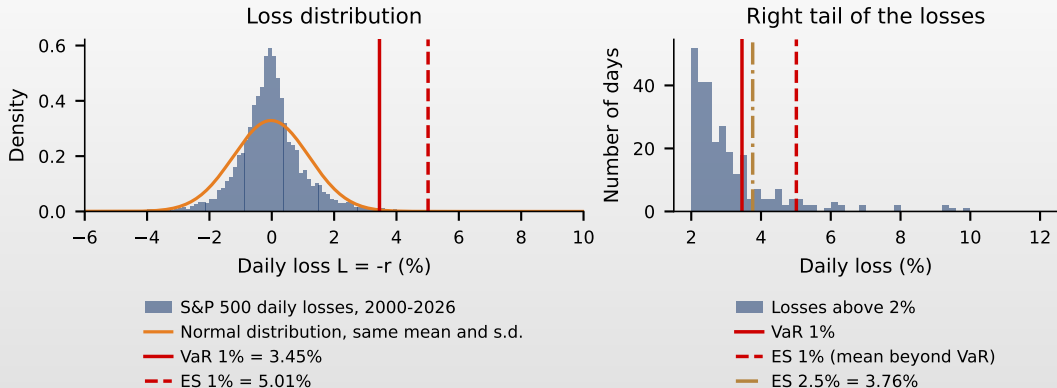
Definitions (1/2): Value-at-Risk

- **VaR** at level α : the loss exceeded with probability α (exactly for continuous L , at most α in general)
 - ▶ $\text{VaR}_\alpha(X) = -\inf\{x \in \mathbb{R} : P(X \leq x) > \alpha\} = -q_\alpha(X)$
 - ▶ In words: VaR is minus the α -quantile of the return, i.e. the return threshold below which only a fraction α of days fall, with the sign changed
- Notation
 - ▶ X : the return of the position over the horizon; $P(X \leq x)$: the probability that it is at most x , i.e. the distribution function at x
 - ▶ $\inf\{\cdot\}$: the infimum (largest lower bound) of the set, here the smallest x at which the distribution function exceeds α
 - ▶ $q_\alpha(X)$: the α -quantile of X , negative for small α ; for a continuous loss $L = -X$: $\text{VaR}_\alpha = q_{1-\alpha}(L)$, the upper quantile of the loss
- Reading of VaR 1%
 - ▶ "With probability 1%, tomorrow's loss will exceed $\text{VaR}_{1\%}$ "
 - ▶ VaR does **not** say how large the loss is beyond this threshold

Definitions (2/2): Expected Shortfall

- **Expected Shortfall:** the average of all quantiles in the tail of probability α
 - ▶ $ES_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du = \frac{1}{\alpha} \int_0^\alpha VaR_u(X) du$
 - ▶ u : the integration variable, a tail probability running from 0 to α ; the integral divided by α is the average of VaR_u over all levels $u \leq \alpha$
- For a continuous distribution: the mean loss on the days beyond VaR
 - ▶ $ES_\alpha(X) = -\mathbb{E}[X \mid X \leq q_\alpha(X)]$; $\mathbb{E}[\cdot \mid \cdot]$: the conditional expectation, here the mean of X over the worst α fraction of outcomes
 - ▶ Always $ES_\alpha \geq VaR_\alpha$: ES looks **into** the tail, VaR only at its entrance; two portfolios with the same VaR can have very different ES
- Other names: Conditional VaR (CVaR), Tail VaR, Average VaR [Rockafellar & Uryasev \(2002\)](#), [Acerbi & Tasche \(2002\)](#)
- Historical estimate of ES: the mean of the observed losses at or above the empirical VaR
 - ▶ A plug-in approximation of the integral, exact when $n\alpha$ is an integer; n : the number of observations

VaR and ES on the S&P 500 (1/2)



VaR and ES on the S&P 500 (2/2)

- Daily losses 2000–2026 against the Normal distribution with the same mean and standard deviation (left); the tail (right)
- Historical $\text{VaR}_{1\%} = 3.45\%$: 68 of 6717 days (1.01%) lost more; on those days the average loss, $\text{ES}_{1\%}$, was 5.01%
- $\text{VaR}_{2.5\%} = 2.51\%$, $\text{ES}_{2.5\%} = 3.76\%$; largest loss 12.77% on 2020-03-16 (COVID-19 crash), about ten standard deviations
- The Normal distribution fits the centre but is far too thin in the tail ([Chapter 1](#))

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The Normal Distribution: ES in One Line (1/2)

- ▣ Setting: $X \sim N(\mu, \sigma^2)$, a Normal return with mean μ and standard deviation σ
 - ▶ Φ, φ : the distribution function and the density of the standard Normal distribution $N(0, 1)$
 - ▶ $z_\alpha = \Phi^{-1}(\alpha) < 0$: the standard Normal α -quantile, e.g. $z_{1\%} = -2.326$
- ▣ VaR: the mean minus $|z_\alpha|$ standard deviations, sign changed
 - ▶ $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha)$
- ▣ ES: the same form, with the larger factor $\varphi(z_\alpha)/\alpha$ in place of $-z_\alpha$
 - ▶ $\text{ES}_\alpha = -\mu - \frac{\sigma}{\alpha} \int_{-\infty}^{z_\alpha} u \varphi(u) du = -\mu + \sigma \frac{\varphi(z_\alpha)}{\alpha}$
 - ▶ The integral is the mean of the standardised shock u over the tail $u \leq z_\alpha$; it equals $-\varphi(z_\alpha)$ because $\varphi'(u) = -u\varphi(u)$

The Normal Distribution: ES in One Line (2/2)

- ▣ Constants to remember
 - ▶ $-z_{1\%} = 2.326$, $\varphi(z_{1\%})/0.01 = 2.665$
 - ▶ $-z_{2.5\%} = 1.960$, $\varphi(z_{2.5\%})/0.025 = 2.338$
 - ▶ So, under the Normal distribution, ES 2.5% is almost equal to VaR 1% (2.338 against 2.326 standard deviations)
- ▣ S&P 500 ($\hat{\mu} = 0.025\%$, $\hat{\sigma} = 1.21\%$; the hat marks a value estimated from the data): Normal $\text{VaR}_{1\%} = 2.80\%$ vs historical 3.45%
 - ▶ Too low by a factor of 1.24
 - ▶ Under the Normal distribution, for a given μ , VaR and ES carry no information beyond σ

Worked Example: the Student- t Distribution

- Model: the return is a shifted and rescaled Student- t variable, $X = m + s T$ with $T \sim t_\nu$
 - ▶ m : location; $s > 0$: scale; ν : degrees of freedom, smaller ν = heavier tails
 - ▶ $t_\alpha = t_\nu^{-1}(\alpha) < 0$: the α -quantile of t_ν ; g_ν : the density of t_ν
- VaR as for the Normal distribution, with t_α in place of z_α ; ES gets an extra factor $(\nu + t_\alpha^2)/(\nu - 1) > 1$ for the heavy tail
 - ▶ $\text{VaR}_\alpha = -(m + s t_\alpha)$, $\text{ES}_\alpha = -m + s \frac{g_\nu(t_\alpha)}{\alpha} \frac{\nu + t_\alpha^2}{\nu - 1}$ (for $\nu > 1$)
 - ▶ Variance $s^2 \nu / (\nu - 2)$ exists only for $\nu > 2$, so s is not the standard deviation
- Maximum likelihood fit to S&P 500 returns: $\hat{\nu} = 2.67$
 - ▶ $\text{VaR}_{1\%} = 3.46\%$ (historical 3.45%), $\text{ES}_{1\%} = 5.71\%$ (historical 5.01%)
- The t distribution fixes the VaR, but with $\hat{\nu} < 3$ it overstates ES: one parameter must fit both the centre and the tail

Coherent Risk Measures (1/2): the Four Axioms

- A **risk measure** ρ maps a loss L (a random variable) to a number, the capital to hold; it is **coherent** if it satisfies four axioms [Artzner, Delbaen, Eber & Heath \(1999\)](#)
 - ▶ L_1, L_2 : two losses; c : a sure amount (constant); $\lambda \geq 0$: a scaling factor (position size)
- **Monotone**: $L_1 \leq L_2 \Rightarrow \rho(L_1) \leq \rho(L_2)$
 - ▶ a position that always loses more needs more capital
- **Translation invariant**: $\rho(L + c) = \rho(L) + c$
 - ▶ adding a sure loss c raises the capital by exactly c
- **Positively homogeneous**: $\rho(\lambda L) = \lambda \rho(L)$
 - ▶ doubling the position doubles the capital
- **Subadditive**: $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$
 - ▶ merging two positions never needs more capital than holding them apart: diversification does not add risk

Coherent Risk Measures (2/2): Why the Axioms Matter

▣ Limits

- ▶ By subadditivity, bank-wide risk does not exceed the sum of the desk risk limits, $\sum_i \rho_i$
- ▶ **No regulatory arbitrage** by splitting a firm

▣ Optimisation: homogeneity + subadditivity $\Rightarrow$ convexity in the portfolio weights

- ▶ Every local minimum is global; uniqueness would need **strict** convexity, which scenario-based ES lacks (it is piecewise linear)

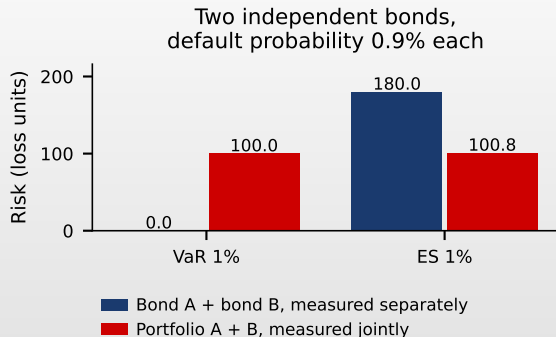
▣ VaR satisfies the first three axioms, but **not** subadditivity in general

- ▶ Exception: jointly elliptical losses (multivariate Normal or Student- t) and $\alpha < 50\%$, where VaR and ES are both $\mu_p + \sigma_p \times \text{constant}$ [McNeil, Frey & Embrechts \(2015\)](#)
- ▶ μ_p, σ_p : mean and standard deviation of the portfolio loss; heavy-tailed refinements in the [Appendix](#)

Counterexample: Two Bonds, Step by Step

- Bonds A and B, independent; each loses 100 with probability $p = 0.9\%$ (default) and 0 otherwise; X : the value change, $X = -L$
- **One bond:** $P(X \leq -100) = p = 0.9\%$, not above 1%, so $q_{1\%} = 0$ and $\text{VaR}_{1\%}(A) = \text{VaR}_{1\%}(B) = 0$
 - ▶ $\text{ES}_{1\%}(A) = (100 \cdot 0.009 + 0 \cdot 0.001)/0.01 = 90$
- **Portfolio A + B:** $P(X \leq -100) = 1 - (1 - p)^2 = 1.79\% > 1\%$, $P(X \leq -200) = p^2 = 0.0081\%$, so $q_{1\%} = -100$
 - ▶ $\text{VaR}_{1\%}(A + B) = 100 > 0 + 0 = \text{VaR}_{1\%}(A) + \text{VaR}_{1\%}(B)$
 - ▶ General formula for a discrete loss $L = -X$: $\text{ES}_\alpha = \frac{1}{\alpha} (\mathbb{E}[L \mathbf{1}\{L > \text{VaR}_\alpha\}] + \text{VaR}_\alpha (\alpha - P(L > \text{VaR}_\alpha)))$
 - ▶ $\mathbf{1}\{\cdot\}$: indicator, 1 if the condition holds and 0 otherwise; first term: losses strictly beyond VaR; second term: the rest of the tail probability α , placed at VaR itself
- $\text{ES}_{1\%}(A + B) = 100.81 \leq 180 = \text{ES}_{1\%}(A) + \text{ES}_{1\%}(B)$

Subadditivity: VaR Fails, ES Holds



- Blue: sum of the two stand-alone measures; red: the measure of the combined portfolio
- Each bond alone “hides” its default in the last 1%; together, at least one default has probability $1.79\% > 1\%$ and VaR jumps to 100
- A VaR limit would reward **concentration**: holding only A looks riskless; ES sees the size of the tail and is coherent [Acerbi & Tasche \(2002\)](#)
- Credit portfolios, options and other skewed pay-offs are where VaR fails in practice

ES as an Optimisation Problem: Rockafellar–Uryasev

- **Theorem Rockafellar & Uryasev (2000):** for any integrable loss L
 - ▶ $ES_\alpha(L) = \min_{v \in \mathbb{R}} F(v)$, $F(v) = v + \frac{1}{\alpha} \mathbb{E}[(L - v)_+]$, $(x)_+ = \max(x, 0)$
 - ▶ v : a candidate threshold; $F(v)$: the threshold plus the expected excess of the loss over it, scaled up by $1/\alpha$
 - ▶ The minimiser is $VaR_\alpha(L)$, the minimum value is $ES_\alpha(L)$; $F(VaR_\alpha)$ is the discrete ES formula of the two-bond slide
- F is convex in v and, for a portfolio loss $-w^\top R$, jointly convex in (w, v)
 - ▶ w : vector of weights; R : vector of asset returns
 - ▶ ES is convex in the weights, which **proves** subadditivity (proof in the [Appendix](#))
 - ▶ With historical scenarios, minimising ES becomes a linear program (the Portfolio VaR section)
- S&P 500 losses, $\alpha = 2.5\%$: $\arg \min F = 2.51\%$ (the minimising v), $\min F = 3.76\%$
 - ▶ Historical VaR and ES (interpolated quantile, tail average): 2.51% and 3.76%
 - ▶ Equal only after rounding: the empirical minimiser is an order statistic

Coherence versus Robustness

- ▣ **Qualitative robustness:** a small change in the data distribution changes the law of the estimator only a little
Cont, Deguest & Scandolo (2010)
 - ▶ Historical VaR is qualitatively robust at distributions with a unique α -quantile; historical ES is **not**: one extreme loss moves it without bound (the precision section)
- ▣ S&P 500, last 500 days: drop the single largest loss (6.16%, 2025-04-04)
 - ▶ $ES_{2.5\%}$: 2.94% $\rightarrow$ 2.60%; $VaR_{1\%}$: 2.73% $\rightarrow$ 2.68%
- ▣ A trade-off, not a free lunch: subadditivity needs the tail beyond the quantile, which is exactly what makes ES sensitive to single observations
- ▣ FRTB chose coherence for capital (ES 2.5%) and kept robust, elicitable VaR for backtesting

Elicitability: Why ES Is Harder to Evaluate (Chapter 8)

- ▣ A risk measure is **elicitable** if it is the minimiser of an expected scoring function
 - ▶ VaR: the quantile $q_\alpha(X)$ minimises the pinball loss $\mathbb{E}[(1\{X \leq v\} - \alpha)(v - X)]$
 - ▶ Scoring function: a penalty computed from the forecast v and the outcome X , lower is better; the pinball loss charges $1 - \alpha$ per unit when X falls below v and α per unit when it stays above
- ▣ ES alone is **not** elicitable [Gneiting \(2011\)](#): no strictly consistent score ranks ES forecasts on their own
- ▣ The pair (VaR, ES) is jointly elicitable [Fissler & Ziegel \(2016\)](#): joint scores such as FZ0 rank (VaR, ES) forecasts ([Appendix](#) ; Chapter 8)
- ▣ Consequence: comparing ES forecasts is harder than comparing VaR forecasts
 - ▶ ES precision is governed by the tail sample $n\alpha$, not by n [Pele & Mazurencu-Marinescu-Pele \(2026\)](#); S&P 500, 26 years: only 168 days in the tail of ES 2.5%

Historical Simulation

- Take the last n daily losses $L_{t-n+1}, \dots, L_t$ and sort them: $L_{(1)} \leq \dots \leq L_{(n)}$; $L_{(i)}$: the i -th smallest loss (the i -th order statistic)
 - ▶ $\widehat{\text{VaR}}_\alpha$ = empirical $(1 - \alpha)$ -quantile, interpolated linearly at position $1 + (n - 1)(1 - \alpha)$ of the sorted losses (the convention used throughout)
 - ▶ $\widehat{\text{ES}}_\alpha$ = mean of the k_α losses $\geq \widehat{\text{VaR}}_\alpha$, $k_\alpha \approx n\alpha$; the exact plug-in of the integral gives the boundary loss the weight $n\alpha - \lfloor n\alpha \rfloor$ ($\lfloor x \rfloor$: the integer part of x), so the two coincide only when $n\alpha$ is an integer (distinct losses)
- Choice of window n : 250 days reacts faster, 1000 days is more precise
 - ▶ With $n = 500$: 5 losses lie beyond VaR 1%; ES 2.5% averages $k = 13$ losses ($n\alpha = 12.5$)
- No loss beyond the worst one in the window can appear in the estimate
- **Ghost effect**: a crash day enters the window, stays for n days, then drops out and VaR falls abruptly

The Cornish–Fisher Expansion (1/2)

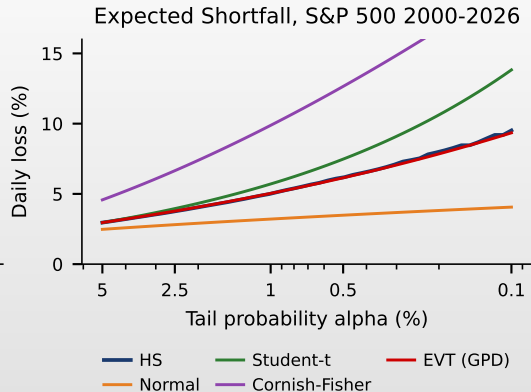
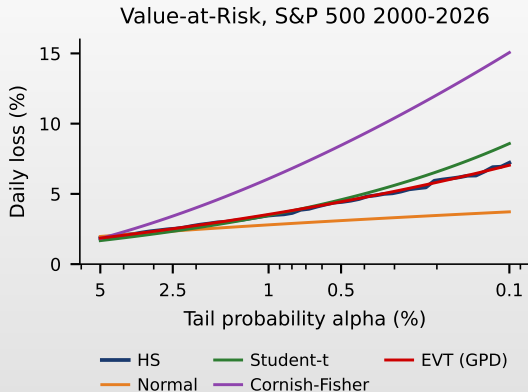
- Adjust the Normal quantile for **skewness** $S = \mathbb{E}[(X - \mu)^3]/\sigma^3$ and **excess kurtosis** $K = \mathbb{E}[(X - \mu)^4]/\sigma^4 - 3$
 Cornish & Fisher (1938)

- $\tilde{z}_\alpha = z_\alpha + \frac{(z_\alpha^2 - 1)S}{6} + \frac{(z_\alpha^3 - 3z_\alpha)K}{24} - \frac{(2z_\alpha^3 - 5z_\alpha)S^2}{36}, \quad \text{VaR}_\alpha = -(\mu + \sigma\tilde{z}_\alpha)$
- $\tilde{z}_\alpha$: the corrected quantile, the Normal quantile $z_\alpha < 0$ plus one term for skewness, one for excess kurtosis and one for squared skewness; $S = K = 0$ gives back the Normal VaR
- $S < 0$ (long left tail of returns) or $K > 0$ (heavy tails) push $\tilde{z}_\alpha$ further below z_α and raise VaR

The Cornish–Fisher Expansion (2/2)

- ▣ ES: average of $-(\mu + \sigma \tilde{z}_u)$ over $u \in (0, \alpha)$, numerically
- ▣ An expansion around the Normal distribution: accurate only for **small** S and K ; for large K the map $z \mapsto \tilde{z}$ is not even monotone (Seminar 7, A6)
- ▣ S&P 500 returns: $S = -0.35$, $K = 10.7$: far outside that range
 - ▶ $\text{VaR}_{1\%}^{CF} = 6.08\%$ versus historical 3.45%: the kurtosis term overshoots

Five Methods Across Tail Probabilities (1/2)

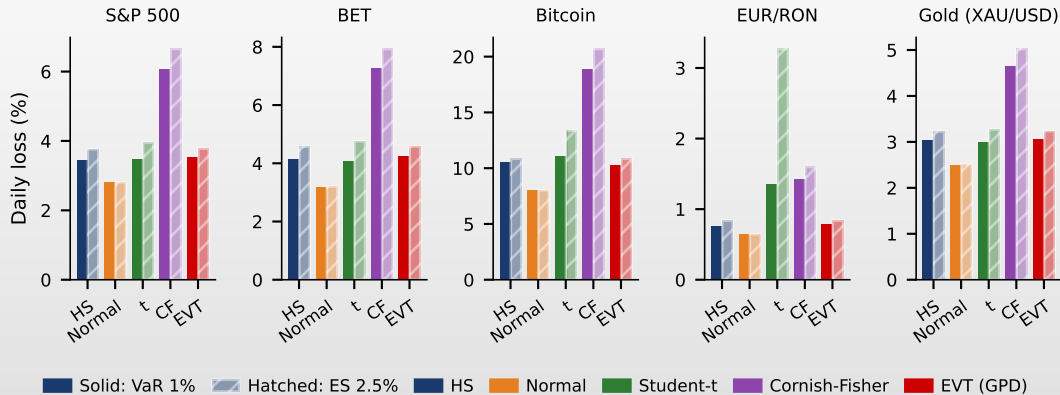


Five Methods Across Tail Probabilities (2/2)

- S&P 500 daily losses 2000–2026, α from 5% down to 0.1%; EVT = generalised Pareto tail, a preview of the Extreme Value Theory section
- VaR 0.1% (one day in four years): historical 7.22%, EVT 7.04%, Student- t 8.58%, Cornish–Fisher 15.06%, Normal 3.72%, about half
- The gap between methods **grows** as α falls, more for ES than for VaR; EVT and historical agree down to 0.5%, below that EVT extrapolates a fitted tail

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Methods Compared on Five Assets (1/2)



Methods Compared on Five Assets (2/2)

- ▣ Solid bars: $\text{VaR}_{1\%}$; hatched bars: $\text{ES}_{2.5\%}$; daily losses in %, full samples (all numbers in the [Appendix](#))
- ▣ Normal: too low everywhere (Bitcoin $\text{VaR}_{1\%}$ 7.99% vs 10.51% historical)
- ▣ Cornish–Fisher: overshoots, since excess kurtosis exceeds 7 in every series (BET 7.25% vs 4.13%)
- ▣ **Exception:** Student- t on EUR/RON
 - ▶ $\hat{\nu} = 1.25$ (infinite variance) and an absurd ES of 3.27%
 - ▶ Many near-zero reference-rate changes drive $\hat{\nu}$ down
- ▣ EVT stays close to historical simulation for VaR 1% in all five series

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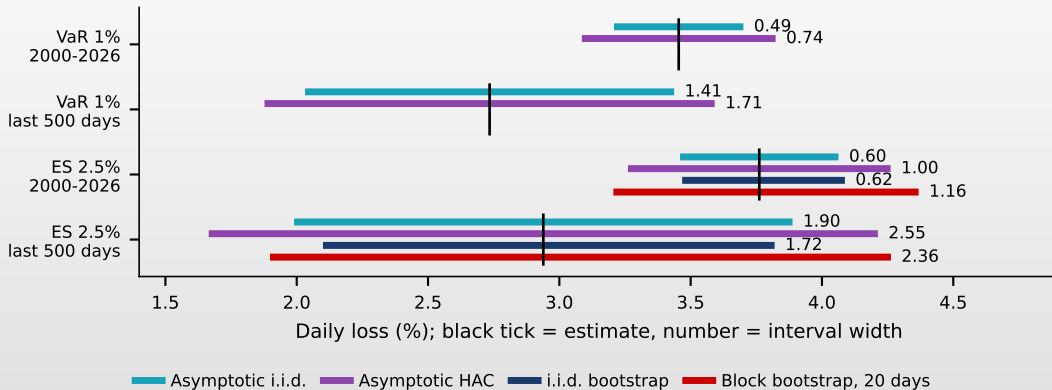
Two Usable Standard Errors (1/2): VaR

- Standard error (SE): the standard deviation of an estimator across repeated samples; a 95% interval is roughly the estimate ± 1.96 SE
- **VaR**: i.i.d. (independent and identically distributed) losses with density f continuous and positive at VaR_α
 - ▶ $\text{SE}(\widehat{\text{VaR}}_\alpha) \approx \frac{1}{f(\text{VaR}_\alpha)} \sqrt{\frac{\alpha(1-\alpha)}{n}}$
 - ▶ $\sqrt{\alpha(1-\alpha)/n}$: the SE of the share of days beyond VaR; dividing by the density $f(\text{VaR}_\alpha)$ turns it into an error on the loss scale
 - ▶ The tail density is small, so the SE is large
 - ▶ $\sigma = 1.2\%$, $n = 500$: SE of $\text{VaR}_{1\%} = 0.20$ percentage points (pp) under the Normal distribution, 0.43 pp under a t_4 with the same σ
- t_4 : the Student- t distribution with 4 degrees of freedom

Two Usable Standard Errors (2/2): ES

- **ES Chen (2008)**: influence function $IF(x) = \text{VaR}_\alpha + (x - \text{VaR}_\alpha)_+/\alpha - \text{ES}_\alpha$, unbounded in x ; if $\mathbb{E}[L^2] < \infty$
 - ▶ $IF(x)$: the effect of one extra observation x on the estimate; it grows without bound with x , so one extreme loss can move ES arbitrarily far
 - ▶ $SE(\widehat{\text{ES}}_\alpha) \approx \frac{1}{\alpha} \sqrt{\frac{\text{Var}((L - \text{VaR}_\alpha)_+)}{n}}$; Var: the variance; estimating VaR adds no first-order term (Rockafellar–Uryasev)
 - ▶ For Pareto-type tails the **relative** error falls like $1/\sqrt{n\alpha}$: the tail count governs precision [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)
- Volatility clustering [Newey & West \(1987\)](#)
 - ▶ Replace the variances by long-run (HAC: heteroskedasticity and autocorrelation consistent) variances of $1\{L_t > \text{VaR}_\alpha\}$ and $(L_t - \text{VaR}_\alpha)_+$
 - ▶ Or use a block bootstrap; derivations in the [Appendix](#)

Precision on the S&P 500 (1/2)



Precision on the S&P 500 (2/2)

- 95% intervals, width in pp next to each bar; bootstrap: 2 000 resamples (Seminar 7, B2); complete table in the [Appendix](#)
- Full sample: clustering widens the ES interval from 0.60 (i.i.d.) to 1.00 (HAC), as the bootstrap does (0.62 i.i.d. vs 1.16 by blocks); last 500 days: ES 2.5% averages 13 losses, HAC interval width 2.55 pp
- These intervals cover the **unconditional** VaR and ES; tomorrow's risk is a conditional question (next section)

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Filtered Historical Simulation (1/2): the Model

- Idea Barone-Adesi, Giannopoulos & Vosper (1999), Hull & White (1998): history gives the **shape** of the shocks, a GARCH model gives today's **scale**
 - ▶ $r_t = \mu_t + \sigma_t z_t$, $\mu_t = c + \phi r_{t-1}$, $\sigma_t^2 = \omega + \alpha_1 (r_{t-1} - \mu_{t-1})^2 + \beta_1 \sigma_{t-1}^2$ (Chapter 5, Bollerslev (1986); α_1 is a GARCH coefficient, not the level α)
- Notation
 - ▶ μ_t : the conditional mean, an AR(1) in the previous return with constant c and coefficient ϕ
 - ▶ σ_t : the conditional volatility; z_t : i.i.d. shock with mean 0 and variance 1, distribution left unspecified
 - ▶ $\omega > 0$: constant; $\alpha_1 \geq 0$: reaction to yesterday's squared surprise; $\beta_1 \geq 0$: weight on yesterday's variance; $\alpha_1 + \beta_1 < 1$: persistence, close to 1 = shocks fade slowly

Filtered Historical Simulation (2/2): Three Steps

- Step 1: estimate the AR(1)-GARCH(1,1) model by quasi-maximum likelihood (Normal likelihood, robust to other shock distributions)
- Step 2: standardised residuals $\hat{z}_t = (r_t - \hat{\mu}_t)/\hat{\sigma}_t$: approximately i.i.d., still heavy-tailed
- Step 3: rescale by tomorrow's volatility
 - ▶ $\text{VaR}_{\alpha,t+1} = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{\xi}_{\alpha}$, $\hat{\xi}_{\alpha} = -q_{\alpha}(\hat{z})$; $\text{ES}_{\alpha,t+1} = -\hat{\mu}_{t+1} - \hat{\sigma}_{t+1} \overline{(\hat{z} \mid \hat{z} \leq q_{\alpha}(\hat{z}))}$
 - ▶ $\hat{\mu}_{t+1}$, $\hat{\sigma}_{t+1}$: the model forecasts for tomorrow; $\hat{\xi}_{\alpha}$: minus the empirical α -quantile of the residuals
 - ▶ $\overline{(\hat{z} \mid \hat{z} \leq q_{\alpha}(\hat{z}))}$: the mean of the residuals at or below that quantile

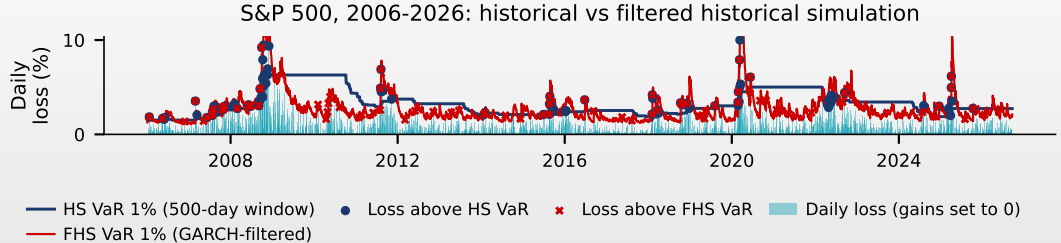
Worked Forecast: S&P 500 on the Next Trading Day

- ▣ AR(1)-GARCH(1,1) on 2000–2026: $\hat{\phi} = -0.053$, $\hat{\alpha}_1 = 0.120$, $\hat{\beta}_1 = 0.860$; persistence $\hat{\alpha}_1 + \hat{\beta}_1 = 0.981$
 - ▶ Forecasts for the first trading day after 18 September 2026: $\hat{\mu}_{t+1} = 0.056\%$, $\hat{\sigma}_{t+1} = 0.725\%$ (sample standard deviation 1.21%: a calm market)
- ▣ Residual quantile $\hat{\xi}_{1\%} = -q_{1\%}(\hat{z}) = 2.79$, above $-z_{1\%} = 2.326$: the filtered shocks are still heavy-tailed
 - ▶ $\text{VaR}_{1\%,t+1} = -0.056 + 0.725 \times 2.791 = 1.97\%$; $\text{ES}_{1\%,t+1} = 2.55\%$
- ▣ Unconditional historical VaR 1%: 3.45%; conditional numbers move with the market, the unconditional one does not
- ▣ Alternatives for the same day: conditional EVT 2.03%, GARCH with Normal shocks 1.63%; all five assets in the
 - ▶ [Appendix](#)

How Uncertain Is Tomorrow's VaR?

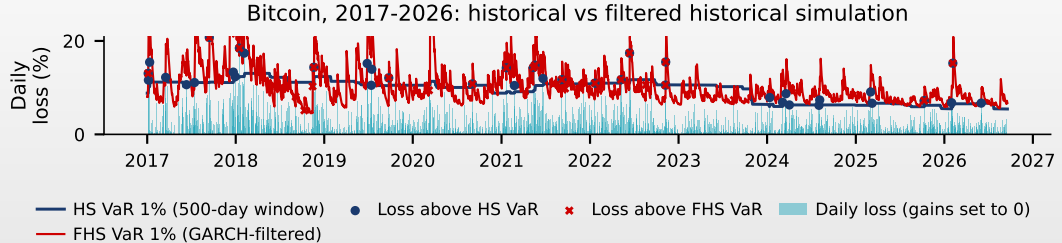
- Two estimated inputs: the GARCH parameters θ (through $\hat{\sigma}_{t+1}$) and the residual quantile $\hat{\xi}_\alpha$ [Francq & Zakoïan \(2015\)](#)
 - ▶ Residual bootstrap [Christoffersen & Gonçalves \(2005\)](#): simulate AR(1)-GARCH(1,1) paths with resampled $\hat{z}$, re-estimate on each path, forecast from the **original** history (algorithm in the [Appendix](#))
- S&P 500, 999 paths: FHS $\text{VaR}_{1\%} = 1.97\%$, 90% interval $[1.90, 2.09]\%$
- The two sources: $\hat{\sigma}_{t+1} = 0.725 [0.708, 0.743]$; $\hat{\xi}_{1\%} = 2.79 [2.68, 2.88]$
 - ▶ Only the quantile varies: $[1.89, 2.03]$; only σ_{t+1} varies: $[1.92, 2.02]$: the residual quantile (about 67 tail residuals) is the larger source
- Conditional EVT (2.03%) lies inside the interval, GARCH with Normal shocks (1.63%) far below it: the choice of tail model matters more than the estimation error of a given model

Historical vs Filtered Historical Simulation: S&P 500



- One-day-ahead $\text{VaR}_{1\%}$; GARCH re-estimated each January on all earlier data, so each forecast uses only information available at its origin
- Days with a loss above VaR 1%, 2006–2026 (5 210 days; target 1%): historical 500 days 1.67%, filtered 1.54%, GARCH with Normal shocks 2.34%
- Historical exceedances **cluster** (2008, 2020): the window reacts months late; filtered VaR peaked at 18.01% on 2020-03-18 and spreads exceedances over time (tests in Chapter 8)

Historical vs Filtered Historical Simulation: Bitcoin

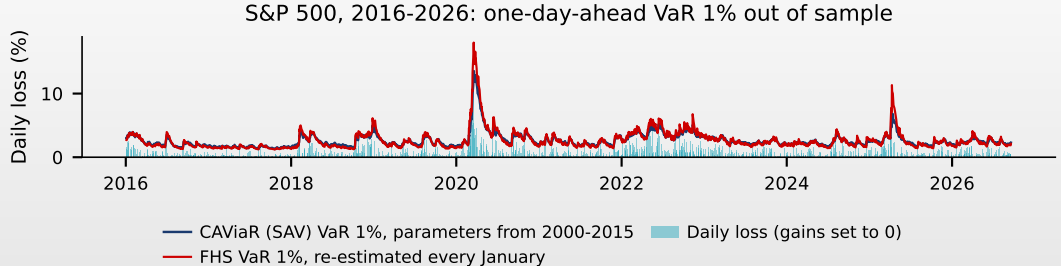


- Same design, forecasts from January 2017, 7 days a week; days with a loss above VaR (3548 days): historical 1.16%, filtered 0.76%, GARCH with Normal shocks 1.58%
- Filtered VaR peaked at 54.83% in March 2020, after the crash of 2020-03-12: a log loss of 46.47%, i.e. a price fall of 37.17% (log and simple losses differ for large moves)
- Volatility has fallen since 2022 and the 500-day window follows with a long delay; crypto as a distinct asset class: [Pele et al. \(2023\)](#)

Modelling the Quantile Directly: CAViaR

- CAViaR (Conditional Autoregressive Value at Risk) Engle & Manganelli (2004): the quantile follows its own recursion; SAV specification
 - ▶ $\text{VaR}_t = \beta_0 + \beta_1 \text{VaR}_{t-1} + \beta_2 |r_{t-1}|$, $q_t = -\text{VaR}_t$
 - ▶ SAV (symmetric absolute value): today's VaR = a constant β_0 + a share $\beta_1 \in (0, 1)$ of yesterday's VaR (persistence) + $\beta_2 > 0$ times yesterday's absolute return
 - ▶ A large move of either sign raises VaR; q_t : the return quantile
- **Quantile regression** Koenker & Bassett (1978): minimise the pinball loss, with no assumption on the shock distribution
 - ▶ $\hat{\beta} = \arg \min_{\beta} \sum_t (\alpha - 1\{r_t < q_t(\beta)\})(r_t - q_t(\beta))$
 - ▶ $\arg \min_{\beta}$: the value of $\beta = (\beta_0, \beta_1, \beta_2)$ that minimises the sum; $q_t(\beta)$: the quantile path implied by β ; elicibility turned into an estimator (optimiser settings in the [Appendix](#))
- S&P 500, $\alpha = 1\%$, estimated on 2000–2015 (4 024 days): $\hat{\beta}_0 = 0.090$, $\hat{\beta}_1 = 0.897$, $\hat{\beta}_2 = 0.246$
 - ▶ Half-life of a shock to VaR: $\ln 0.5 / \ln \hat{\beta}_1 = 6.4$ days
- CAViaR reacts linearly to $|r_{t-1}|$, GARCH to r_{t-1}^2

CAViaR vs Filtered Historical Simulation, Out of Sample



- S&P 500, 2016–2026 (2 693 days): CAViaR with parameters fixed on 2000–2015; FHS re-estimated each January (a different refitting design)
- Exceedances of VaR 1%: CAViaR 1.30% (35), FHS 1.30% (35), historical 500 days 1.41% (38); mean pinball loss (lower is better): 0.039, 0.037, 0.052
- COVID-19, 20 February – 30 April 2020 (50 days): exceedances 6 / 3 / 9, peak VaR 13.5% / 18.0% / 4.5%: after a crash FHS jumps higher
- Counts alone do not establish that one model is better: significance is a [Chapter 8](#) question

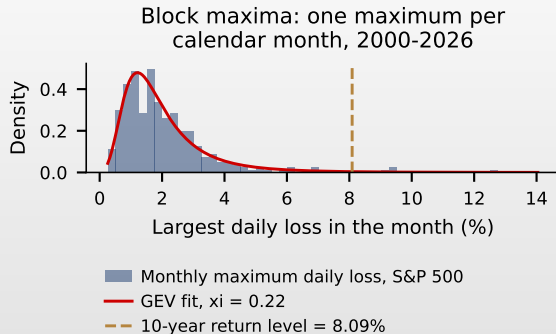
Block Maxima and the GEV Distribution (1/2)

- Split the sample into blocks of n days; keep each block maximum $M_n = \max(L_1, \dots, L_n)$
 - ▶ Fisher–Tippett–Gnedenko theorem [Fisher & Tippett \(1928\)](#), [Gnedenko \(1943\)](#): if $(M_n - b_n)/a_n$ has a non-degenerate limit for i.i.d. losses, the limit is GEV
 - ▶ $a_n > 0$, b_n : normalising constants that rescale and recentre the maximum; GEV: generalised extreme value distribution
- The GEV distribution function: the probability that a block maximum is at most x
 - ▶ $H_\xi(x) = \exp \left\{ - \left(1 + \xi \frac{x-\mu}{\sigma} \right)^{-1/\xi} \right\}$, $1 + \xi(x - \mu)/\sigma > 0$ ($\xi = 0$: $\exp\{-e^{-(x-\mu)/\sigma}\}$)
 - ▶ μ : location (where the maxima sit); $\sigma > 0$: scale (their spread); ξ : shape (the tail type, next slide)

Block Maxima and the GEV Distribution (2/2)

- The **shape** ξ decides the tail
 - ▶ $\xi > 0$: Fréchet, power-law tail with tail index $1/\xi$ (financial returns)
 - ▶ $\xi = 0$: Gumbel domain, exponential-type tails (e.g. Normal, lognormal); $\xi < 0$: Weibull, bounded tail
- **Return level** for k blocks: the level exceeded on average once every k blocks, $H_\xi^{-1}(1 - 1/k)$, with H_ξ^{-1} the quantile function of the GEV
- In finance we model **only the tail** Embrechts, Klüppelberg & Mikosch (1997), Longin (2000)

Monthly Maxima of S&P 500 Losses



- 321 calendar months, 2000–2026; GEV by maximum likelihood
- $\hat{\xi} = 0.22$ (Fréchet), $\hat{\mu} = 1.37$, $\hat{\sigma} = 0.79$: a power-law tail
- Return levels: 10 years 8.09%, 30 years 10.95%; largest monthly maximum 12.77%
- Monthly maxima do not give a daily VaR directly: the conversion assumes independent days, which clustering violates ([Appendix](#))
- One value per month wastes data: the next method uses all large losses

Peaks over Threshold and the GPD (1/2)

- Fix a high threshold u ; model the **excesses** $Y = L - u$ given $L > u$
 - ▶ Pickands–Balkema–de Haan theorem [Pickands \(1975\)](#), [Balkema & de Haan \(1974\)](#): for a wide class of distributions, as u approaches the right end-point of the distribution the excesses follow a generalised Pareto distribution
- The GPD distribution function: the probability that an excess is at most y
 - ▶ $G_{\xi, \beta}(y) = 1 - (1 + \xi y / \beta)^{-1/\xi}$ ($\xi \neq 0$), $1 - e^{-y/\beta}$ ($\xi = 0$)
 - ▶ $y \geq 0$, and $y < -\beta/\xi$ if $\xi < 0$; scale $\beta > 0$: the typical size of an excess; shape ξ : the same as in the GEV, $\xi > 0$ = heavy tail

Peaks over Threshold and the GPD (2/2)

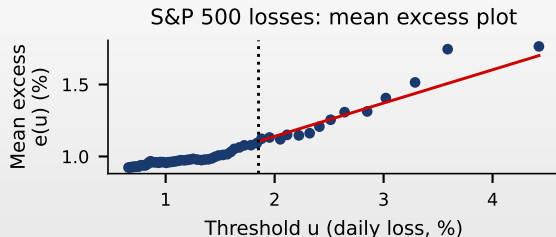
- Tail estimator with N_u exceedances among n observations [McNeil, Frey & Embrechts \(2015\)](#)

- ▶ $\widehat{\text{VaR}}_\alpha = u + \frac{\hat{\beta}}{\hat{\xi}} \left[\left(\frac{n}{N_u} \alpha \right)^{-\hat{\xi}} - 1 \right], \quad \widehat{\text{ES}}_\alpha = \frac{\widehat{\text{VaR}}_\alpha}{1 - \hat{\xi}} + \frac{\hat{\beta} - \hat{\xi}u}{1 - \hat{\xi}} \quad (\alpha < N_u/n, \hat{\xi} \neq 0; \text{ES needs } \hat{\xi} < 1)$
- ▶ In words: VaR is the threshold u plus the GPD quantile of the excess that leaves probability α in the tail; N_u/n is the share of days above u
- ▶ ES adds to VaR the mean excess beyond VaR, which grows with $\hat{\xi}$; $\hat{\xi} \geq 1$ means an infinite mean
- Bias–variance trade-off: a low u violates the limit theorem, a high u leaves few exceedances

Choosing the Threshold: the Mean Excess Function

- ▣ **Mean excess function:** $e(u) = \mathbb{E}[L - u \mid L > u]$; empirically, the average excess above u , plotted against u
- ▣ For a GPD tail $e(u)$ is **linear** in u
 - ▶ $e(u) = \frac{\beta + \xi(u - u_0)}{1 - \xi}$ for $u \geq u_0$, finite only if $\xi < 1$; slope $\xi/(1 - \xi)$: upward for heavy tails, flat for exponential tails
 - ▶ u_0 : the lowest threshold above which the GPD holds; β : the GPD scale at u_0
- ▣ Rule of thumb: choose u where the plot becomes roughly linear; check the stability of $\hat{\xi}$ across thresholds
- ▣ The Hill estimator of [Chapter 1 Hill \(1975\)](#) estimates the same tail index $1/\xi$

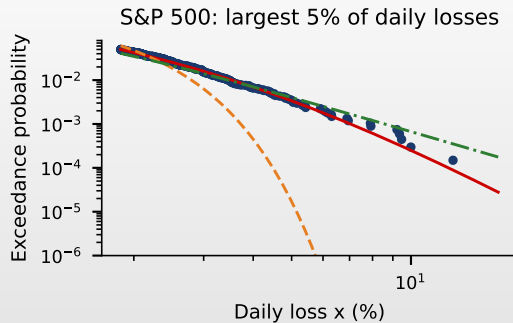
Mean Excess Plot: S&P 500



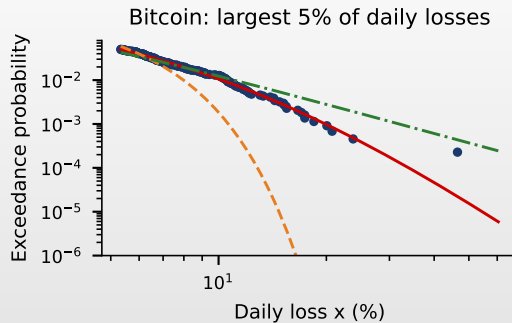
- Empirical mean excess $e(u)$
- GPD implied: $(\beta + \xi(u - u_0)) / (1 - \xi)$
- Threshold $u_0 = 1.85\%$ (top 5% of losses above it)

- Thresholds with 20% down to 0.5% of daily losses above them; red line: GPD fitted to the largest 5%
- The mean excess rises with u : a heavy (Fréchet-type) tail, not the flat line of an exponential tail
- Selected $u_0 = 1.85\%$ leaves $N_u = 336$ exceedances; mean excess 1.09%, GPD slope $\hat{\xi}/(1 - \hat{\xi}) = 0.23$
- Beyond VaR 1% the points scatter: each averages very few losses

GPD Tails: S&P 500 and Bitcoin (1/2)



- Empirical tail $P(L \geq x)$
- Normal distribution
- GPD, $\xi = 0.19$
- Student-t, $\nu = 2.7$



- Empirical tail $P(L \geq x)$
- Normal distribution
- GPD, $\xi = 0.16$
- Student-t, $\nu = 2.2$

GPD Tails: S&P 500 and Bitcoin (2/2)

- Log-log scale: probability of a daily loss above x , largest 5% of losses; the Normal tail falls far below the data, the Student- t tail is close to the GPD
- S&P 500: $\hat{\xi} = 0.19$ (SE 0.065), tail index 5.3, $\text{VaR}_{1\%} = 3.54\%$; Bitcoin: $\hat{\xi} = 0.16$ (SE 0.078), $\text{VaR}_{1\%} = 10.23\%$; all five series in the [Appendix](#)
- $\text{SE}(\hat{\xi}) = (1 + \hat{\xi})/\sqrt{N_u}$ assumes independent exceedances; clustering makes it too small (next slide)

 MFM_ch7_gpd_tail

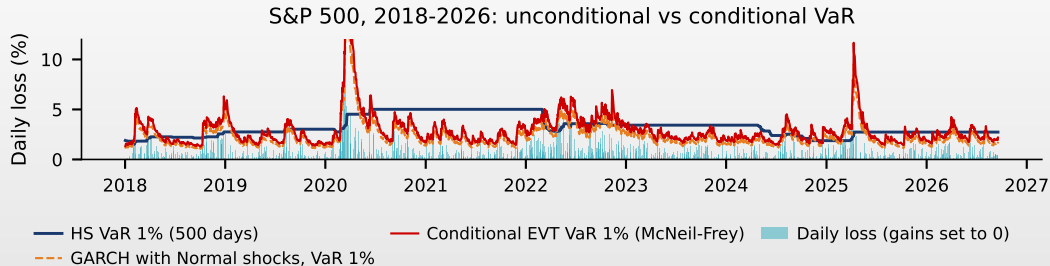
Clustered Extremes: the Extremal Index

- Stationary losses with clustering: if $nP(L > u_n) \rightarrow \tau$, then $P(M_n \leq u_n) \rightarrow e^{-\theta\tau}$, $\theta \in (0, 1]$ [Embrechts, Klüppelberg & Mikosch \(1997\)](#)
 - ▶ u_n : a threshold rising with n ; τ : the expected number of exceedances of u_n in n days; M_n : the maximum of the n losses
 - ▶ θ = extremal index, $1/\theta$ = mean cluster size; i.i.d. data: $\theta = 1$; n dependent days act like $n\theta$ independent ones
- S&P 500, GPD threshold $u = 1.85\%$ ($N = 336$ exceedances): intervals estimator [Ferro & Segers \(2003\)](#) gives $\hat{\theta} = 0.19$, i.e. 62 clusters of 5.4 exceedances on average
 - ▶ GPD on cluster maxima: $\hat{\xi} = 0.152$ (SE 0.146) vs 0.188 (SE 0.065) on all exceedances: still positive, but its 95% interval now includes 0
- Standardised GARCH residuals: $\hat{\theta} = 0.99$: filtering removes the clustering, which is why conditional EVT fits the GPD after the filter
- Estimator, declustering rule and the block-maxima correction: [Appendix](#)

Conditional EVT (McNeil and Frey)

- ▣ McNeil & Frey (2000): the FHS pipeline, with one step changed
- ▣ Filter: AR(1)-GARCH(1,1) by quasi-maximum likelihood, residuals $\hat{z}_t$, forecasts $\hat{\mu}_{t+1}$, $\hat{\sigma}_{t+1}$ (as in FHS)
- ▣ **Tail step**: instead of the empirical quantile of $-\hat{z}$, fit a GPD to its largest 10% and take its VaR $\hat{z}_\alpha^{GPD}$ and ES $\hat{e}_\alpha^{GPD}$
 - ▶ $\text{VaR}_{\alpha,t+1} = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1}\hat{z}_\alpha^{GPD}$, $\text{ES}_{\alpha,t+1} = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1}\hat{e}_\alpha^{GPD}$
- ▣ A smooth parametric tail beyond the data: matters most far in the tail (VaR 0.5%, VaR 0.1%), where only a handful of residuals exist

Conditional EVT on the S&P 500



- One-day-ahead $\text{VaR}_{1\%}$, 2018–2026, parameters re-estimated each January; exceedances (2 190 days; target 1%): conditional EVT 1.23%, GARCH with Normal shocks 2.42%, historical 500 days 1.60%

- GARCH–Normal has the right **timing** but the wrong **level**; conditional EVT keeps the timing and fixes the level: the tail of the shocks matters, not only their scale

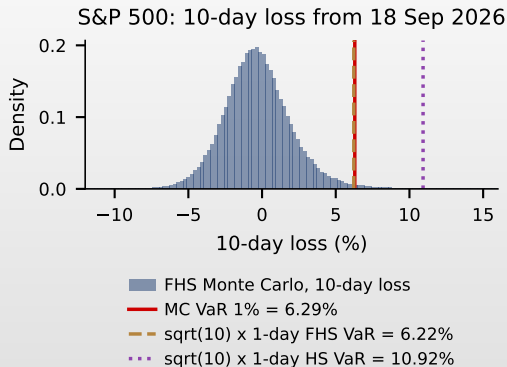
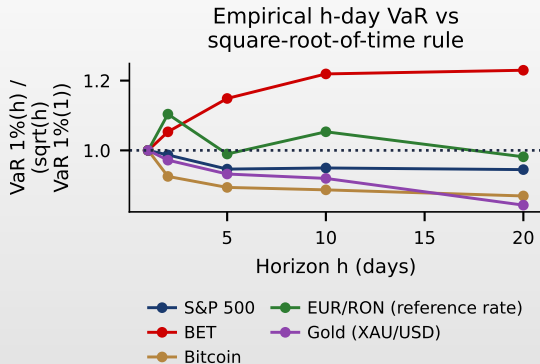
The Square-Root-of-Time Rule: Assumptions and Failures

- i.i.d. $N(0, \sigma^2)$ daily losses: $\text{VaR}_\alpha^{(h)} = \sqrt{h} \text{VaR}_\alpha^{(1)}$, $\text{ES}_\alpha^{(h)} = \sqrt{h} \text{ES}_\alpha^{(1)}$; Basel 1996 allowed $\sqrt{10} = 3.162$ [Basel Committee \(1996\)](#)
 - ▶ $\text{VaR}_\alpha^{(h)}$, $\text{ES}_\alpha^{(h)}$: the measures for the loss summed over h days; the standard deviation of a sum of h i.i.d. terms grows like $\sqrt{h}$
 - ▶ Needs zero mean, no autocorrelation, constant volatility and Normal shocks; each fails in the data ([Chapters 1, 2, 5](#))
- **Autocorrelation** (variance ratio, [Chapter 2](#))
 - ▶ $\text{Var}(r_{t+1} + \dots + r_{t+h}) = h\sigma^2 \text{VR}(h)$, with $\text{VR}(h)$ the variance of the h -day sum divided by h times the daily variance
 - ▶ $\text{VR} = 1$ without autocorrelation; $\text{VR} > 1$ with positive autocorrelation, and then $\sqrt{h}$ **underestimates** the h -day risk
- **Volatility clustering**: volatility reverts to its mean over h days
 - ▶ Calm today $\Rightarrow \sqrt{h}$ too low; turbulent today $\Rightarrow$ too high
- **Heavy tails**: for i.i.d. shocks with tail index $a > 2$ the far-tail quantile of a sum grows like $h^{1/a} < \sqrt{h}$ [Daníelsson \(2011\)](#)
 - ▶ a : tail index, $P(L > x)$ decays like x^{-a} ; smaller a = heavier tail
 - ▶ With jumps the rule can badly misstate multi-day risk [Daníelsson & Zigrand \(2006\)](#)

Filtered Monte Carlo over h Days

- Simulate many future paths from the fitted model and read VaR and ES from the simulated losses: handles all three failures at once, and non-linear pay-offs (options)
- For each path $j = 1, \dots, M$ (M : the number of simulated paths)
 - ▶ draw $z_{t+1}^{(j)}, \dots, z_{t+h}^{(j)}$ with replacement from the standardised residuals $\hat{z}$
 - ▶ update $\sigma^{(j)}$ and $r^{(j)}$ day by day with the AR(1)-GARCH(1,1) recursion, starting from today's state
 - ▶ aggregate: $L^{(j)} = -\sum_{i=1}^h r_{t+i}^{(j)}$, the h -day loss of path j (i : day within the path)
- $\widehat{\text{VaR}}_\alpha, \widehat{\text{ES}}_\alpha$ = historical estimates on the M simulated losses; simulation error falls as $1/\sqrt{M}$

Square-Root-of-Time in the Data (1/2)



Square-Root-of-Time in the Data (2/2)

- Left: empirical $\text{VaR}_{1\%}$ of overlapping h -day losses divided by $\sqrt{h}$ times the one-day value (overlapping sums: far fewer independent observations); right: filtered Monte Carlo, 200 000 paths
- Ratio at $h = 10$: BET 1.22 (lag-1 autocorrelation 0.109) and EUR/RON 1.05: $\sqrt{h}$ too low; S&P 500 0.95, gold 0.92, Bitcoin 0.89: conservative
- S&P 500 10-day $\text{VaR}_{1\%}$: filtered Monte Carlo 6.29%, $\sqrt{10} \times$ filtered one-day 6.22%, $\sqrt{10} \times$ historical one-day 10.92%: the big gap is conditional vs unconditional

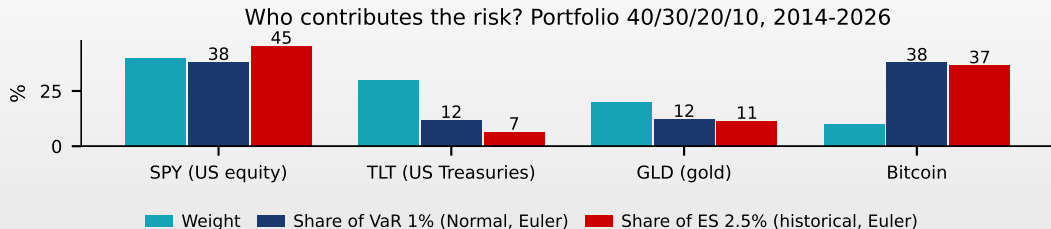
Portfolio VaR and Euler Allocation (1/2)

- Setting: portfolio loss $L_p = -w^\top R$
 - ▶ w : vector of weights; R : vector of simple returns (simple returns add across assets), with covariance matrix Σ ; σ_i : standard deviation of asset i
- Normal returns, mean ignored: VaR is a multiple of the portfolio standard deviation
 - ▶ $\text{VaR}_\alpha(L_p) = z_{1-\alpha}\sigma_p$, $\sigma_p = \sqrt{w^\top \Sigma w}$; $z_{1-\alpha} = -z_\alpha > 0$
 - ▶ For long-only $w \geq 0$, the undiversified VaR $\sum_i z_{1-\alpha} w_i \sigma_i$ (the sum of stand-alone VaRs) is larger unless all correlations are 1

Portfolio VaR and Euler Allocation (2/2)

- **Marginal VaR:** the change in portfolio VaR per unit increase of weight w_i
 - ▶ $\partial \text{VaR}_\alpha / \partial w_i = z_{1-\alpha} (\Sigma w)_i / \sigma_p$; $(\Sigma w)_i$: the covariance of asset i with the portfolio
- **Component VaR** $C_i = w_i \partial \text{VaR}_\alpha / \partial w_i$: the part of VaR attributed to asset i
 - ▶ Euler's theorem for a function homogeneous of degree 1: $\sum_i w_i \partial \rho / \partial w_i = \rho$, so the components add up, $\sum_i C_i = \text{VaR}_\alpha$ [Tasche \(2008\)](#)
- ES, no distributional assumption (continuous losses): the mean loss of asset i on the portfolio's tail days
 - ▶ $C_i^{ES} = \mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}_\alpha(L_p)]$, $\sum_i C_i^{ES} = \text{ES}_\alpha(L_p)$
 - ▶ Historical estimate: the average over the tail days; a negative C_i means the asset **hedges** the portfolio

Who Contributes the Risk?



- SPY (S&P 500 ETF) 40%, TLT (long-term US Treasury ETF) 30%, GLD (gold ETF) 20%, Bitcoin 10%, rebalanced daily; simple returns on 3017 common days, 2014–2026; teal: weights, blue: shares of Normal $\text{VaR}_{1\%}$, red: shares of historical $\text{ES}_{2.5\%}$

- $\text{VaR}_{1\%} = 1.77\%$ vs 3.11% undiversified (-43%); $\text{ES}_{2.5\%} = 2.15\%$ vs 3.87% (-44%)

- Bitcoin: 10% of the money, 37.8% of VaR and 36.6% of ES; TLT: 30% of the money, only 6.5% of ES (it lost 0.47% on the portfolio's worst days)

Minimum-ES Portfolio: a Linear Program

- With N historical scenarios R_s , the Rockafellar–Uryasev objective becomes a **linear program** Rockafellar & Uryasev (2000)
 - ▶ $\min_{w, v, u} v + \frac{1}{\alpha N} \sum_s u_s$ subject to $u_s \geq -w^\top R_s - v$, $u_s \geq 0$, $\sum_i w_i = 1$, $w \geq 0$
 - ▶ $s = 1, \dots, N$: the scenarios; u_s : an auxiliary variable equal, at the optimum, to the excess $(-w^\top R_s - v)_+$ of scenario s over the threshold v ; the optimal v is the portfolio VaR
 - ▶ Constraints: fully invested ($\sum_i w_i = 1$), no short sales ($w \geq 0$)
- Same four assets, 3 017 common days, $\alpha = 2.5\%$: weights 31% / 44% / 25% / 0%
 - ▶ $\text{ES}_{2.5\%} = 1.67\%$, against 2.15% for the 40/30/20/10 portfolio
- Long-only minimum variance: 35% / 43% / 22% / 0%, $\text{ES}_{2.5\%} = 1.67\%$: for these assets the two criteria almost agree; they part ways for options and credit
- Several portfolios can attain the same minimum: report the solution together with its tail scenarios

Liquidity-Adjusted VaR

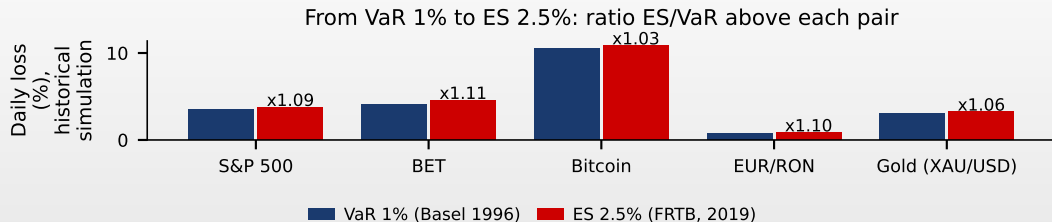
- VaR assumes positions can be sold at the mid price; in reality we pay half the bid–ask spread
 - ▶ **Relative spread** $S = (\text{ask} - \text{bid})/\text{mid}$; it widens exactly when markets fall
- **Bangia, Diebold, Schuermann & Stroughair (2001)**: $\text{LVaR}_\alpha = \text{VaR}_\alpha + \frac{1}{2}(\mu_S + a\sigma_S)$
 - ▶ μ_S, σ_S : mean and standard deviation of the spread; a : a multiplier (here S is the spread, not skewness)
 - ▶ VaR plus the cost of crossing half a stressed spread, a standard deviations above its mean
- Banca Transilvania, 1 million RON, last two years: $\text{VaR}_{1\%} = 3.16\%$; with assumed $\mu_S = 0.5\%$, $\sigma_S = 0.3\%$, $a = 3$: $\text{LVaR} = 3.86\%$ (+22%)
- Large positions also move the price (endogenous liquidity): the regulator handles this with longer horizons

From Basel 1996 to FRTB (2019)

	Basel 1996 Basel Committee (1996)	FRTB 2019 Basel Committee (2019)
Risk measure	VaR 1%, computed daily	ES 2.5%, computed daily for every trading desk
Horizon	10 days; $\sqrt{10}$ scaling allowed	10-day base, scaled by liquidity horizons (large-cap equity 10, small-cap 20, credit spreads up to 120 days)
Data	at least one year of history	calibrated to the most severe 12-month stress period
Capital	$\max(\text{yesterday's VaR}, \text{multiplier} \geq 3 \times 60\text{-day average VaR})$; the multiplier rises by 0–1 after poor backtests (traffic-light zones, Chapter 8)	multiplier $m_c = 1.5$, increased after poor backtests
Backtesting	VaR 1% exceptions (Chapter 8)	one-day VaR 1% and VaR 2.5%: VaR tests, ES sets the capital

- Critique after 2008: VaR ignores the size of tail losses and one-year windows forget crises; EU application date of FRTB own-funds requirements: 1 January 2027 [Delegated Regulation \(EU\) 2025/1496](#)

VaR 1% vs ES 2.5% on Five Assets



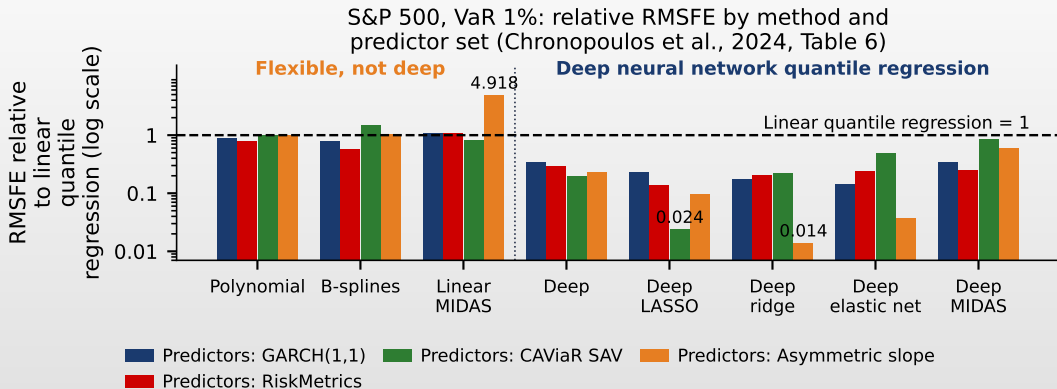
- Historical simulation, daily losses, full samples; number above each pair: $ES_{2.5\%}/VaR_{1\%}$
- Normal benchmark: $2.338/2.326 \approx 1.005$, the same capital for thin tails; Student- t : 1.01 at $\nu = 30$, 1.07 at $\nu = 4$
- Data: S&P 500 1.09, BET 1.11, Bitcoin 1.03, EUR/RON 1.10, gold 1.06: the switch from VaR 1% to ES 2.5% raises the one-day measure by 3–11%
- Warning: $ES_{\alpha} \geq VaR_{\alpha}$ holds at the **same** α ; across levels heavy tails alone do not guarantee it
- Stressed calibration and liquidity horizons add more: Seminar 7, C2

Case Study (1/4): Question and Data

- **Question** Chronopoulos, Raftapostolos & Kapetanios (2024) (*Journal of Financial Econometrics*)
 - ▶ Does VaR forecasting improve if the conditional quantile may depend on its predictors in an **unknown non-linear** way, learned by a deep neural network?
- Data (paper, Section 3): S&P 500 daily log returns, September 1985 – August 2020; $\alpha = 1\%, 5\%, 10\%$
 - ▶ Predictors x_t : GARCH or RiskMetrics volatility, CAViaR state variables
 - ▶ Last 2000 days out of sample, one fixed tuning window; details in the [Appendix](#)
- Benchmark: linear quantile regression Koenker & Bassett (1978) on the same predictors
 - ▶ i.e. CAViaR-type linearity (the Conditional VaR and ES section)

Case Study (2/4): the Model

- Model (Eqs. 5–10): the conditional return quantile is a neural network of the predictors
 - ▶ $Q_{r_t}(\alpha | x_t) \approx G_\alpha(x_t, w)$
 - ▶ $Q_{r_t}(\alpha | x_t)$: the α -quantile of r_t given x_t ; G_α : a feed-forward network with ReLU hidden layers ($\max(\cdot, 0)$); w : its weights
- Estimation: minimise the pinball loss plus a penalty on w (none, LASSO, ridge or elastic net); w^* : the fitted weights
- Target: the daily return quantile 10 days ahead (Eq. 23)
 - ▶ $\widehat{\text{VaR}}_{t+10|\mathcal{F}_t} = -G_\alpha(x_t, w^*)$; $\mathcal{F}_t$: the information available at day t
- Evaluation: RMSFE (root mean squared forecast error) relative to the benchmark; below 1 = smaller error than linear quantile regression
 - ▶ Significance: Diebold–Mariano (DM) test of equal forecast accuracy

Case Study (3/4): Relative Forecast Error, $\alpha = 1\%$ 

Case Study (4/4): Relative Forecast Error, $\alpha = 1\%$

- Chronopoulos, Raftapostolos & Kapetanios (2024), Table 6: RMSFE relative to linear quantile regression (dashed line = 1), log scale; S&P 500, last 2 000 days to August 2020
- Flexible non-network models: 7 of 12 bars at or above 1; all 16 deep-network bars and the 4 deep MIDAS bars below 1 (0.014–0.850), all significant in the DM test except deep MIDAS for SAV
- Caution: RMSFE is minimised by the conditional **mean**, not the quantile; quantile accuracy needs the pinball loss
- One fixed tuning window: whether a 98% cut survives rolling re-estimation is open (Seminar 7, C3)

 MFM_ch7_crk_table6

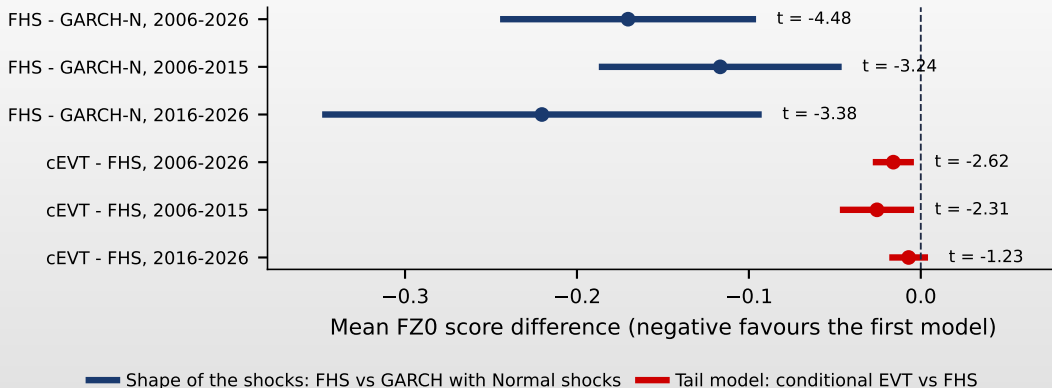
An Open Question and What to Check (1/2)

- Can daily data rank competing (VaR 1%, ES 1%) models? [Nolde & Ziegel \(2017\)](#), [Patton, Ziegel & Chen \(2019\)](#), [Bayer & Dimitriadis \(2022\)](#), [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)
 - ▶ Obstacles: ES alone is not elicitable, and only about 1% of days are informative
- FZ0 score: a joint penalty for a (VaR, ES) forecast given the realised return, lower is better [Patton, Ziegel & Chen \(2019\)](#)
 - ▶ $S(v, e; Y) = -\frac{1}{\alpha e} 1\{Y \leq v\}(v - Y) + \frac{v}{e} + \ln(-e) - 1$
 - ▶ Y : the realised return; $v_t = -\text{VaR}_{\alpha,t}$ and $e_t = -\text{ES}_{\alpha,t}$: the forecast quantile and tail mean, both negative; the first term penalises returns below v
- Test: $d_t = S_t^A - S_t^B$, the daily score difference of models A and B
 - ▶ $H_0: \mathbb{E}[d_t] = 0$ (equal accuracy) vs $H_1: \mathbb{E}[d_t] \neq 0$; Diebold–Mariano t -statistic with HAC errors, approximately $N(0, 1)$ under H_0 ; $t < -1.96$: A significantly better at 5%
 - ▶ One interval per pre-registered subperiod

An Open Question and What to Check (2/2)

- An AI assistant can draft the literature map, the code and a referee report; the researcher checks
 - ▶ sign and level: $\text{VaR } 1\% = -q_{1\%}$, positive for a loss (recompute one day of FZ0 step by step)
 - ▶ look-ahead bias: each forecast uses only data before its day
 - ▶ scoring: joint (VaR, ES) scores only, never the MSE (mean squared error) of ES alone [Gneiting \(2011\)](#)
 - ▶ power: count the tail days
 - ▶ every DOI (Digital Object Identifier) resolved; reusable prompts in the [Appendix](#)

Mini-Case: Can We Rank ES Models? (1/2)



Mini-Case: Can We Rank ES Models? (2/2)

- Claim tested: “conditional EVT is the best (VaR 1%, ES 1%) model for the S&P 500”; replicate first: FHS exceedance rate 1.54% (80 of 5 210 days, 2006–2026)
- Mean FZ0 (lower is better): conditional EVT 1.267, FHS 1.283, GARCH–Normal 1.453; bars: HAC 95% intervals for $\mathbb{E}[d_t]$
- The shape of the shocks matters robustly: Normal shocks lose in both halves ($t = -3.24, -3.38$)
- The EVT gain over FHS is ten times smaller and not significant in 2016–2026 ($[-0.018, 0.004]$): the gain is not demonstrated, and this does not prove the two models are equivalent

 [MFM_ch7_ai_discovery_case](#)

Your Turn: A Team-Project Seed

- Question: does a dynamic semiparametric (VaR, ES) model [Patton, Ziegel & Chen \(2019\)](#) beat FHS and conditional EVT at ES 2.5%, out of sample?
 - ▶ Markets: the S&P 500, the BET and Bitcoin
- Replicate first: the S&P 500 FHS VaR 1% of 1.97% and the FZ0 ranking of the mini-case
- Deliverables
 - ▶ A one-page pre-registration: models, levels, scores, subperiods, power (tail days) and what would falsify the claim
 - ▶ Code on the daily market data of the course, a table of all comparisons with adjusted p -values, a referee-style self-critique
- Log prompts and accepted outputs in `AI_USE.md`, every error caught (flipped sign, wrong level label, look-ahead bias, invented DOI) in `AI_ERRORS.md`; code: [Quantlet](#)

Which Method When?

Situation	Method	Fails when
Quick check, long stable history	Historical simulation, 500–1000 days	volatility changes; nothing beyond the worst loss
Daily risk after a volatility shock	FHS or conditional EVT, with a bootstrap interval	the GARCH model is misspecified
VaR 0.5% and deeper, stress tests	Peaks over threshold (GPD), conditional or not	threshold badly chosen; clustered exceedances
Multi-day horizon, options	Filtered Monte Carlo	model risk dominates
Risk budgeting across desks or assets	Euler contributions to ES	few tail days behind each contribution
Regulatory capital	ES 2.5%, 10 days, stressed period, VaR backtests	the stress window is short and noisy

- Every VaR or ES is an estimate: report it with its tail count and an interval

Check Yourself, and Next: Chapter 8

Retrieval questions

- ▣ Why can VaR reward concentration?
- ▣ Why can ES not reward it?
- ▣ Why does the precision of ES depend on $n\alpha$ rather than on n ?
- ▣ When is $\sqrt{10}$ scaling too low?
- ▣ When is it too high?

Chapter 8: backtesting

- ▣ How often did losses exceed VaR?
- ▣ Were the exceedances independent?
- ▣ Coverage tests and the Basel traffic-light zones
- ▣ Comparing (VaR, ES) forecasts with joint scores Fissler & Ziegel (2016)
- ▣ Further reading: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017), Čížek, Härdle & Weron (2011), McNeil, Frey & Embrechts (2015)

Appendix: Data Used in This Chapter

Series	Type	Period	Days
S&P 500	US equity index	2000–2026	6 717
BET	BVB equity index	2000–2026	6 670
Bitcoin	crypto-asset, 7 days a week	2014–2026	4 384
EUR/RON	BNR reference rate	2005–2026	5 352
Gold	USD per troy ounce, weekdays	2000–2026	6 956
Portfolio	SPY, TLT, GLD (ETFs) and Bitcoin	2014–2026	3 017
Banca Transilvania	BVB share (liquidity example)	2024–2026	496

- Daily closing values up to 18 September 2026; daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$
- Each series on its own calendar; the portfolio uses the days on which all four prices exist
- ETFs: prices adjusted for dividends; SPY tracks the S&P 500, TLT long US Treasury bonds, GLD gold

Appendix: Proof of the Rockafellar–Uryasev Representation

- **Proof:** $F(v) = v + \mathbb{E}[(L - v)_+]/\alpha$ is convex, with right derivative (slope just to the right of v) $1 - P(L > v)/\alpha$
 - ▶ It is ≥ 0 exactly when $P(L > v) \leq \alpha$, i.e. $v \geq \text{VaR}_\alpha$; $F(\text{VaR}_\alpha)$ is the discrete ES formula of the two-bond slide
- **Corollary:** $(w, v) \mapsto v + \mathbb{E}[(-w^\top R - v)_+]/\alpha$ is jointly convex; minimising over v keeps convexity
 - ▶ ES is convex in w ; with positive homogeneity this proves subadditivity
- Envelope argument: $\partial F/\partial v = 0$ at $v = \text{VaR}_\alpha$, so a small error in VaR changes the minimum only to second order: the source of the ES standard error without a VaR term

Appendix: When Is VaR Subadditive After All?

- **Jointly elliptical** losses (multivariate Normal, multivariate Student- t) and $\alpha < 0.5$: VaR is subadditive [McNeil, Frey & Embrechts \(2015\)](#)
 - ▶ There VaR and ES are both $\mu_p + \sigma_p \times \text{constant}$, with $\sigma_p \leq \sum_i \sigma_i$; Normal or t marginals alone are not enough
- **Heavy but not too heavy tails**: for multivariate regularly varying losses with tail index above 1, VaR is subadditive far enough in the tail (asymptotically, as $\alpha \rightarrow 0$) [Danielsson et al. \(2013\)](#)
 - ▶ Estimated tail indices of equity and FX returns are in that range (between 4.8 and 6.5 in the Extreme Value Theory section), but the result is asymptotic, not a guarantee at a given α
- It fails for discrete jumps (defaults), extremely heavy tails and some option pay-offs
- VaR vs ES in practice: ES needs more data for the same precision [Yamai & Yoshida \(2005\)](#)

Appendix: VaR Aggregation under Dependence Uncertainty

- Margins $F_1, \dots, F_d$ known, copula unknown (Chapter 6): $\text{VaR}_\alpha(\sum_i L_i)$ lies in an interval Embrechts, Puccetti & Rüschendorf (2013)
 - ▶ F_i : distribution function of the loss L_i of asset i , d : number of assets; copula: the dependence structure that joins the margins
 - ▶ The comonotone value $\sum_i \text{VaR}_\alpha(L_i)$ (all losses moving perfectly together) is **not** the upper end: VaR is not subadditive
- **Rearrangement algorithm**: discretise the tail beyond each $\text{VaR}_\alpha(L_i)$ in N points (one column per asset)
 - ▶ Order each column opposite to the sum of the others until the row sums are flat; worst VaR $\approx$ smallest row sum
- 40/30/20/10 portfolio of the Portfolio VaR section, daily $\text{VaR}_{1\%}$, empirical margins, $N = 2\,000$
 - ▶ Historical dependence 1.95%; comonotone 3.63%; worst case 4.76% ($1.31 \times$ comonotone)
- Capital based on VaR carries dependence model risk of the same order as the diversification benefit

Appendix: Entropy and Market Risk

- **Differential entropy** of a density f : $H(L) = - \int f(x) \ln f(x) dx$, a measure of dispersion without moments
 - ▶ Larger H = more spread-out density; $\ln$: natural logarithm
 - ▶ For a location-scale family $L = \mu + \sigma Z$ (Z : the standardised loss): $H(L) = \ln \sigma + H(Z)$, so $\text{VaR}_\alpha = \mu + e^{H(L)-H(Z)} q_{1-\alpha}(Z)$: entropy plays the role of the scale
- **Pele, Lazar & Dufour (2017)**: the entropy of intraday returns is linked to intraday VaR and ES and helps forecast daily VaR (EUR/JPY)
- **Pele & Mazurencu-Marinescu-Pele (2019)**: high-frequency entropy gives the best daily VaR forecasts for Bitcoin among the models compared
- **Pele, Lazar & Mazurencu-Marinescu-Pele (2019)**: a nonparametric tail-entropy measure as an alternative estimator of ES

Appendix: VaR and ES by Method and Asset

	VaR _{1%} (%)					ES _{2.5%} (%)				
	Hist.	Normal	<i>t</i>	CF	EVT	Hist.	Normal	<i>t</i>	CF	EVT
S&P 500	3.45	2.80	3.46	6.08	3.54	3.76	2.81	3.95	6.65	3.78
BET	4.13	3.19	4.06	7.25	4.22	4.57	3.21	4.76	7.93	4.58
Bitcoin	10.51	7.99	11.06	18.89	10.23	10.84	8.03	13.37	20.69	10.84
EUR/RON	0.76	0.64	1.36	1.43	0.78	0.84	0.65	3.27	1.60	0.84
Gold	3.04	2.50	3.00	4.64	3.07	3.23	2.51	3.27	5.02	3.23

- Daily losses, full samples; CF = Cornish–Fisher, EVT = generalised Pareto tail fitted to the largest 5% of losses (the Extreme Value Theory section)
- Kurtosis: S&P 500 10.7, BET 11.1, Bitcoin 12.0, EUR/RON 15.1, gold 7.7; Bitcoin needs about three times the S&P 500 buffer, EUR/RON about a fifth

Appendix: Asymptotic Theory for Historical VaR and ES

- ▣ **VaR:** for i.i.d. losses with density f continuous and positive at VaR_α ,

$$\sqrt{n}(\widehat{\text{VaR}}_\alpha - \text{VaR}_\alpha) \rightarrow N\left(0, \frac{\alpha(1-\alpha)}{f(\text{VaR}_\alpha)^2}\right)$$
 - ▶ $\rightarrow$: convergence in distribution as $n \rightarrow \infty$; the variance of the limit is the asymptotic variance (avar), and $\text{SE} = \sqrt{\text{avar}/n}$
- ▣ **ES** [Chen \(2008\)](#): influence function $\text{IF}(x) = \text{VaR}_\alpha + (x - \text{VaR}_\alpha)_+/\alpha - \text{ES}_\alpha$
 - ▶ No term from estimating VaR: $\partial F/\partial v = 0$ at $v = \text{VaR}_\alpha$; if $\mathbb{E}[(L - \text{VaR}_\alpha)_+]^2 < \infty$, the variance of the limit of $\sqrt{n}(\widehat{\text{ES}}_\alpha - \text{ES}_\alpha)$ is $\text{avar} = \text{Var}((L - \text{VaR}_\alpha)_+)/\alpha^2$
 - ▶ Pareto-type tail with index $a > 2$, $\alpha \rightarrow 0$: $\text{Var}((L - \text{VaR}_\alpha)_+) \approx \frac{2\alpha \text{VaR}_\alpha^2}{(a-1)(a-2)}$, so $\text{SE} \propto \text{VaR}_\alpha/\sqrt{n\alpha}$ ($\propto$: proportional to) [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)
- ▣ Stationary, weakly dependent losses: replace the variances by long-run (HAC) variances of $1\{L_t > \text{VaR}_\alpha\}$ and $(L_t - \text{VaR}_\alpha)_+$ [Newey & West \(1987\)](#), or use a block bootstrap

Appendix: Precision on the S&P 500, Complete Table

	n	$\widehat{\text{VaR}}_{1\%}$	SE i.i.d.	SE HAC	$\widehat{\text{ES}}_{2.5\%}$	SE i.i.d.	SE HAC
2000–2026	6 717	3.45	0.13	0.19	3.76	0.15	0.26
Last 500 days	500	2.73	0.36	0.44	2.94	0.48	0.65

- Density $f(\text{VaR})$ by a Gaussian kernel (Silverman bandwidth); HAC: Bartlett kernel, lag $\lfloor 4(n/100)^{2/9} \rfloor$ (10 and 5) [Newey & West \(1994\)](#)
- Full sample, $n = 6\,717$: SE of $\text{VaR}_{1\%}$ under the Normal distribution with $\sigma = 1.2\%$ would be 0.055, under t_4 0.12; clustering inflates the SE by 50–70%

Appendix: GARCH Filters and Tomorrow's Volatility

	$\hat{\phi}$	$\hat{\alpha}_1$	$\hat{\beta}_1$	$\hat{\alpha}_1 + \hat{\beta}_1$	$\hat{\sigma}$ (sample)	$\hat{\sigma}_{t+1}$	$-q_{1\%}(\hat{z})$
S&P 500	-0.053	0.120	0.860	0.981	1.21	0.73	2.79
BET	0.104	0.207	0.787	0.994	1.40	1.76	2.74
Bitcoin	-0.009	0.127	0.843	0.970	3.49	3.00	2.88
EUR/RON	0.169	0.200	0.780	0.980	0.28	0.11	2.74
Gold	-0.022	0.046	0.943	0.989	1.09	1.43	2.68

- $\hat{\sigma}_{t+1}$: forecast for the first trading day after 18 September 2026 (19 September for Bitcoin), in %
- Persistence close to 1: volatility shocks die out slowly ([Chapter 5](#)); S&P 500 and EUR/RON are calm today, BET and gold turbulent
- $-q_{1\%}(\hat{z})$ between 2.68 and 2.88, above $-z_{1\%} = 2.326$: the filtered shocks are still heavy-tailed

Appendix: Conditional VaR and ES for the Next Trading Day

	VaR _{1%} (%)				ES _{1%} (%)			
	GARCH	FHS	c-EVT	Hist.	GARCH	FHS	c-EVT	Hist.
S&P 500	1.63	1.97	2.03	3.45	1.88	2.55	2.56	5.01
BET	3.80	4.53	4.63	4.13	4.39	6.16	6.20	6.28
Bitcoin	6.89	8.56	8.68	10.51	7.90	12.22	12.19	14.29
EUR/RON	0.23	0.27	0.26	0.76	0.27	0.34	0.35	1.12
Gold	3.32	3.82	3.86	3.04	3.80	4.94	4.93	4.21

- GARCH: GARCH with Normal shocks; c-EVT: conditional extreme value theory (the Extreme Value Theory section); Hist.: unconditional, full sample
- GARCH with Normal shocks is the lowest of the three conditional methods in every row: volatility alone does not create the tail

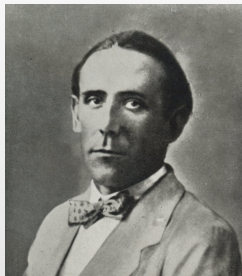
Appendix: Estimation Risk in Conditional VaR

- $\text{VaR}_{\alpha,t+1} = -\mu_{t+1}(\theta) + \sigma_{t+1}(\theta) \xi_{\alpha}$, $\xi_{\alpha} = -q_{\alpha}(z)$: **both** θ (GARCH, by QML) and ξ_{α} (a residual quantile) are estimated
 - ▶ θ : the AR(1)-GARCH(1,1) parameters; QML: quasi-maximum likelihood; $\mu_{t+1}(\theta)$, $\sigma_{t+1}(\theta)$: the forecasts they imply
- [Francq & Zakoïan \(2015\)](#): estimate the **risk parameter** (θ, ξ_{α}) jointly; under strict stationarity, $\mathbb{E}[z_t^4] < \infty$ and a positive density of z at $q_{\alpha}(z)$, the conditional VaR is asymptotically Normal, with a variance from $\hat{\theta}$, from $\hat{\xi}_{\alpha}$ and from their covariance
- **Residual bootstrap** [Christoffersen & Gonçalves \(2005\)](#), for $b = 1, \dots, B$ (B : number of bootstrap samples; a star marks a value computed on a bootstrap sample)
 - ▶ simulate an AR(1)-GARCH(1,1) path of the sample length with resampled $\hat{z}$ and re-estimate θ^* on it
 - ▶ σ_{t+1}^* from the **original** returns with θ^* ; ξ_{α}^* from the path's own residuals; $\text{VaR}_{t+1}^* = -\mu_{t+1}^* + \sigma_{t+1}^* \xi_{\alpha}^*$; percentile interval: the 5% and 95% quantiles of the B values VaR_{t+1}^*
- Heavy tails and skewness of z slow the convergence; [Spierdijk \(2016\)](#) compares interval methods for ARMA–GARCH VaR in exactly this case

Appendix: CAViaR Estimation Settings and Joint (VaR, ES) Models

- ▣ Estimation settings of [Engle & Manganelli \(2004\)](#): 10^4 random starts in $(0, 1)^3$, the best 10 refined by simplex and quasi-Newton steps; the recursion starts at $\text{VaR}_1 = -\hat{q}_\alpha$, minus the empirical α -quantile of the first 300 returns; these days stay in the estimation sample
- ▣ The pair (VaR, ES) can be **estimated** by minimising a strictly consistent joint score [Fissler & Ziegel \(2016\)](#); [Chapter 8](#) uses the same scores to **evaluate** forecasts
 - ▶ [Patton, Ziegel & Chen \(2019\)](#): dynamic semiparametric (VaR, ES) models, e.g. with GAS dynamics, estimated by minimising the FZ0 loss; consistency and asymptotic normality
 - ▶ [Taylor \(2019\)](#): CAViaR-type recursions for VaR and ES, estimated by an asymmetric Laplace likelihood
 - ▶ [Dimitriadis & Bayer \(2019\)](#): linear regression of the quantile and of ES on covariates (realised and implied volatility), with standard errors
- ▣ Common thread: the scoring function that ranks forecasts is also the objective function that estimates the model

Appendix: Emil Gumbel and the Statistics of Extremes



Emil Julius Gumbel (1891–1966)

Photo: unknown author (c. 1931); public domain;
[Wikimedia Commons](#)

- Fisher & Tippett (1928): the maximum of many i.i.d. variables, suitably rescaled, can have only three limit forms
- Gnedenko (1943) proved the result rigorously
- Gumbel (1958) wrote the first textbook and applied it to floods, droughts and life lengths
 - ▶ The Gumbel distribution is one of the three limit forms



Appendix: From Monthly Maxima to a Daily VaR

- ▣ Independent days: $P(M_n \leq x) \approx P(L \leq x)^n$, so $\text{VaR}_\alpha \approx H^{-1}((1 - \alpha)^n)$ with $n = 20.9$ trading days per month and H^{-1} the quantile function of the fitted GEV
 - ▶ $\text{VaR}_{1\%} \approx H^{-1}(0.810) = 2.83\%$ vs historical 3.45%; $\text{VaR}_{0.1\%} \approx 6.19\%$
- ▣ Volatility clustering breaks the independence assumption, and the approximation is too low here
- ▣ With the extremal index: $P(M_n \leq x) \approx F(x)^{n\hat{\theta}}$, F : the distribution function of the daily loss
 - ▶ $\hat{\theta} = 0.35$ (threshold at VaR 1%): $\text{VaR}_{1\%} \approx H^{-1}(0.929) = 4.14\%$ instead of 2.83%; with $\hat{\theta} = 0.19$: 5.09%
 - ▶ The correction moves the answer by more than the error it fixes: the block-maxima route to a daily VaR is fragile

Appendix: GPD Fits for Five Series

	u (%)	N_u	$\hat{\xi}$	$SE(\hat{\xi})$	$1/\hat{\xi}$	$VaR_{1\%}$	$ES_{1\%}$
S&P 500	1.85	336	0.19	0.065	5.3	3.54	5.04
BET	1.93	334	0.21	0.066	4.8	4.22	6.34
Bitcoin	5.33	220	0.16	0.078	6.1	10.23	14.36
EUR/RON	0.38	268	0.18	0.072	5.5	0.78	1.14
Gold	1.68	348	0.15	0.062	6.5	3.07	4.22

- $\hat{\xi}$ between 0.15 and 0.21 for all five series, each more than two i.i.d. standard errors above 0: heavy tails everywhere (clustering makes these SEs too small)
- Tail index $1/\hat{\xi}$ between 4.8 and 6.5: the fitted upper tails have finite fourth moments, but every 95% interval for ξ reaches above 0.25, so finite fourth moments cannot be confirmed (compare [Chapter 1](#))

Appendix: The Extremal Index: Estimator and Declustering

- ▣ **Intervals estimator** Ferro & Segers (2003) from the N exceedance times of u and the gaps T_i between them
 - ▶ $\hat{\theta} = \min \left\{ 1, \frac{2(\sum_i (T_i - 1))^2}{(N - 1) \sum_i (T_i - 1)(T_i - 2)} \right\}$ if $\max T_i > 2$ (else with T_i, T_i^2); it compares the squared mean gap with the mean squared gap: short gaps inside clusters push $\hat{\theta}$ below 1
- ▣ **Declustering** Ferro & Segers (2003): target $C = \lfloor \hat{\theta} N \rfloor + 1$ clusters, separated by the $C - 1$ largest gaps; fit the GPD to the cluster maxima
 - ▶ Ties: if the $(C - 1)$ -th and C -th largest gaps are equal, C is reduced until they differ, so the realised number of clusters can be below the target
- ▣ S&P 500: target $C = 63$, 62 clusters after the tie rule; at the historical VaR 1% ($u = 3.45\%$, $N = 68$): $\hat{\theta} = 0.35$
 - ▶ The i.i.d. SE of $\hat{\xi}$ was less than half the SE from the cluster maxima

Appendix: The Portfolio: Volatilities, Correlations and Contributions

	Weight	σ_i (%)	corr. SPY	corr. TLT	corr. GLD
SPY	40%	1.10	1.00		
TLT	30%	0.93	-0.17	1.00	
GLD	20%	1.03	0.07	0.25	1.00
Bitcoin	10%	4.14	0.23	-0.02	0.09

	σ_i (%)	Marginal VaR	C_i	share	C_i^{ES}	share
SPY	1.10	1.68	0.67	38.1%	0.98	45.4%
TLT	0.93	0.70	0.21	11.8%	0.14	6.5%
GLD	1.03	1.08	0.22	12.3%	0.25	11.4%
Bitcoin	4.14	6.68	0.67	37.8%	0.79	36.6%
Total	$\sigma_p = 0.76$		1.77	100%	2.15	100%

- SPY carries more of the ES than of the Normal VaR (45.4% vs 38.1%): its losses concentrate on the portfolio's worst days more than covariances imply; calling this tail dependence needs a separate analysis ([Chapter 6](#))

Appendix: The Basel Committee and the EU Timetable



The BIS tower in Basel, seat of the Basel Committee on Banking Supervision

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- The Basel Committee on Banking Supervision sets global capital standards; national supervisors (in the EU, via the Capital Requirements Regulation) turn them into law
- **1996:** market risk enters the capital rules; banks may use internal VaR models [Basel Committee \(1996\)](#)
- **2019:** the Fundamental Review of the Trading Book standard replaces VaR by ES in the internal-models approach [Basel Committee \(2019\)](#); each jurisdiction sets its own application date
- EU: [Delegated Regulation \(EU\) 2025/1496](#) postponed the FRTB own-funds requirements to 1 January 2027; until then banks apply the earlier VaR-based framework

Appendix: Case Study, Data and Method in Detail (1/2)

- **Data** (paper, Section 3.1): S&P 500 daily log returns $r_t = \ln(P_t/P_{t-1})$, September 1985 – August 2020, $T = 9\,053$; $\alpha = 1\%$, 5% and 10%
 - ▶ Four predictor sets x_t (Eqs. 14–17, following [Giacomini & Komunjer \(2005\)](#)): GARCH(1,1) σ_t ; RiskMetrics σ_t with decay factor $\lambda = 0.94$; CAViaR SAV ($\text{VaR}_{t-1}, |r_{t-1}|$); CAViaR asymmetric slope ($\text{VaR}_{t-1}, r_{t-1}^+, r_{t-1}^-$), $r^+ = \max(r, 0)$, $r^- = \max(-r, 0)$
 - ▶ The paper writes τ for our α ; $\text{VaR}_{\alpha,t} = -Q_{r_t}(\alpha \mid x_t)$, as in the VaR and Expected Shortfall section
- **Design** (paper, Section 3.2): the last 2 000 days are out of sample; the first 7 053 tune the model once (fixed forecast window)
 - ▶ Time-series CV (cross-validation, Eq. 12): gap $\delta = 5$ days around each left-out point, $\kappa = 20$ random test points; depth and width in $\{1, 5, 10\}$, learning rate and the penalty weight λ (not the RiskMetrics λ) five values in $[0.001, 0.01]$, batch size $\{10, 20\}$, dropout up to 20%, Adam optimiser (stochastic gradient descent with adaptive step sizes), early stopping (paper, Section 1.4)

Appendix: Case Study, Data and Method in Detail (2/2)

- **Rivals:** linear quantile regression (the benchmark), quantile polynomial and B-splines, linear MIDAS (Eq. 18) and a deep MIDAS network
 - ▶ **Evaluation** (paper, Section 3.3): RMSFE relative to the linear benchmark; DM test [Diebold & Mariano \(1995\)](#) on quantile losses with the correction of [Harvey, Leybourne & Newbold \(1997\)](#); CQFE (conditional quantile forecast encompassing) test [Giacomini & Komunjer \(2005\)](#): does one forecast already contain the information of the other; smoothing $\eta = 0.005$
 - ▶ **Motivation:** [Patton, Ziegel & Chen \(2019\)](#) extended CAViaR to (VaR, ES) with parametric dynamics; this paper keeps VaR alone but drops linearity

Appendix: Case Study, Complete Results (Tables 6 and 7)

$\alpha = 1\%$	Polyn.	Splines	MIDAS	Deep	LASSO	Ridge	Elnet	Deep MIDAS
GARCH	0.909	0.791	1.083	0.346	0.236	0.177	0.146	0.346
RiskMetrics	0.780	0.570	1.066	0.289	0.141	0.207	0.239	0.246
SAV	1.000	1.511	0.838	0.198	0.024	0.225	0.498	0.850
Asym. slope	1.000	1.040	4.918	0.235	0.096	0.014	0.037	0.601

- RMSFE relative to linear quantile regression, $\alpha = 1\%$ (Table 6; LASSO, Ridge, Elnet = penalised deep networks): all 16 deep entries are below 1 and significant at 1% in the DM test
 - ▶ Over all levels, the deep networks cut the error by 50%–98%, the deep MIDAS by 11%–84%; the best penalty changes with the predictor set (elastic net for GARCH, LASSO for RiskMetrics and SAV, ridge for the asymmetric slope)
- In most cases the linear benchmark has smaller errors than the polynomial and spline regressions (ratios up to 1.716 at other α levels in Table 6); linear MIDAS reaches 4.918; the results do not isolate the contributions of architecture, penalisation and CV
- CQFE at $\alpha = 1\%$ (Table 7): the non-linear block encompasses its rivals 757 times (42 losses, 751 inconclusive); the linear block 173 times (15, 170); many inconclusive pairs mean that combining forecasts could still add information

Appendix: Case Study, Replication and Limitations

- **Replicate** on the daily market data of the course: the S&P 500 series starts in January 1990, so 1985–1989 is missing (a data limit)
 - ▶ Keep the evaluation window of the paper (the last 2 000 returns up to 31 August 2020, from 19 September 2012): 5 726 tuning days instead of 7 053
 - ▶ Build the GARCH and SAV predictors, fit linear and deep quantile regressions at $\alpha = 1\%$ and recompute these two rows of Table 6 (Seminar 7, C3)
- **Limits stated by the authors**
 - ▶ One fixed estimation window: recursive or rolling re-tuning over 2 000 days is computationally challenging (paper, Section 3.2)
 - ▶ Networks overfit easily, hence penalties, dropout, CV and early stopping (paper, Section 1.4); the output is read only through SHAP values and partial derivatives (paper, Section 4)
- **Open extensions** (Seminar 7, C3)
 - ▶ Does the gain survive rolling re-estimation on 2020–2026 and the BET?
 - ▶ Can a network trained on the FZ0 loss of [Patton, Ziegel & Chen \(2019\)](#) forecast (VaR, ES) jointly, as FRTB needs?

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List papers since 2015 on comparative backtesting of (VaR, ES) with strictly consistent scoring functions: DOI, scoring function, test, sample size and power findings."*
 - ▶ Resolve every DOI on Crossref; check the sign convention of the score in the paper
- **Hypotheses:** *"For FHS, conditional EVT and GARCH with Normal shocks, state which features of the tail each captures and what each predicts for the FZ0 differences in calm and in turbulent years."*
- **Data and code:** *"Write Python that computes the FZ0 loss for daily VaR 1% and ES 1% forecasts on the S&P 500 from the daily market data of the course, with the sign convention $\text{VaR} = -q_\alpha$, and a Diebold–Mariano test with HAC errors."*
 - ▶ Replicate first: the FHS exceedance rate 1.54% over 5 210 days (the Conditional VaR and ES section)

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness:** *“Write a pre-registered plan: subperiods 2006–2015 and 2016–2026, levels 1% and 2.5%, the FZ0 and a second consistent score, HAC lags 5 and 20, a second index.”*
 - ▶ Size check: simulate from a fitted GARCH-t and score two forecasts with equal expected FZ0; the test must reject in about 5% of simulations
- **Critique:** *“Act as a referee: is my ranking driven by 2008 and 2020, by the score’s sign convention or by overlapping estimation windows? Give the test that would expose each.”*
- **Report:** every model pair, level and period; the number of tail days behind each comparison
- **Multiple testing:** models $\times$ levels $\times$ periods $\times$ assets: fix the family in advance and adjust the p -values; an unresolvable DOI or a misattributed score: drop the claim and log it in `AI_ERRORS.md`
- Fix the plan before the first run: any specification chosen after seeing results is exploratory and is reported as such

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Measuring Statistical Risk
- ☐  XFG Advanced Methods in Quantitative Finance

Code and data

- ☐ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_07



References (1/6)

- ▣ Acerbi, C., Tasche, D. (2002). [On the coherence of expected shortfall](#). *Journal of Banking & Finance*, 26(7), 1487–1503.
- ▣ Artzner, P., Delbaen, F., Eber, J.-M., Heath, D. (1999). [Coherent measures of risk](#). *Mathematical Finance*, 9(3), 203–228.
- ▣ Balkema, A.A., de Haan, L. (1974). [Residual life time at great age](#). *Annals of Probability*, 2(5), 792–804.
- ▣ Bangia, A., Diebold, F.X., Schuermann, T., Stroughair, J.D. (2001). [Modeling liquidity risk, with implications for traditional market risk measurement and management](#). In *Risk Management: The State of the Art*, Springer, 3–13.
- ▣ Barone-Adesi, G., Giannopoulos, K., Vosper, L. (1999). [VaR without correlations for portfolios of derivative securities](#). *Journal of Futures Markets*, 19(5), 583–602.
- ▣ Basel Committee on Banking Supervision (1996). [Amendment to the capital accord to incorporate market risks](#). Bank for International Settlements.
- ▣ Basel Committee on Banking Supervision (2019). [Minimum capital requirements for market risk](#). Bank for International Settlements.
- ▣ Bayer, S., Dimitriadis, T. (2022). [Regression-based expected shortfall backtesting](#). *Journal of Financial Econometrics*, 20(3), 437–471.
- ▣ Bollerslev, T. (1986). [Generalized autoregressive conditional heteroskedasticity](#). *Journal of Econometrics*, 31(3), 307–327.
- ▣ Chen, S.X. (2008). [Nonparametric estimation of expected shortfall](#). *Journal of Financial Econometrics*, 6(1), 87–107.

References (2/6)

- ▣ Christoffersen, P., Gonçalves, S. (2005). Estimation risk in financial risk management. *Journal of Risk*, 7(3), 1–28.
- ▣ Chronopoulos, I., Raftapostolos, A., Kapetanios, G. (2024). Forecasting Value-at-Risk using deep neural network quantile regression. *Journal of Financial Econometrics*, 22(3), 636–669.
- ▣ Čížek, P., Härdle, W.K., Weron, R. (Eds.) (2011). *Statistical Tools for Finance and Insurance*, 2nd ed. Springer.
- ▣ Cont, R., Deguest, R., Scandolo, G. (2010). Robustness and sensitivity analysis of risk measurement procedures. *Quantitative Finance*, 10(6), 593–606.
- ▣ Cornish, E.A., Fisher, R.A. (1938). Moments and cumulants in the specification of distributions. *Revue de l'Institut International de Statistique*, 5(4), 307–320.
- ▣ Daníelsson, J. (2011). *Financial Risk Forecasting*. Wiley.
- ▣ Daníelsson, J., Jorgensen, B.N., Samorodnitsky, G., Sarma, M., de Vries, C.G. (2013). Fat tails, VaR and subadditivity. *Journal of Econometrics*, 172(2), 283–291.
- ▣ Daníelsson, J., Zigrand, J.-P. (2006). On time-scaling of risk and the square-root-of-time rule. *Journal of Banking & Finance*, 30(10), 2701–2713.
- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Dimitriadis, T., Bayer, S. (2019). A joint quantile and expected shortfall regression framework. *Electronic Journal of Statistics*, 13(1), 1823–1871.

References (3/6)

- ▣ Embrechts, P., Klüppelberg, C., Mikosch, T. (1997). *Modelling Extremal Events for Insurance and Finance*. Springer.
- ▣ Embrechts, P., Puccetti, G., Rüschendorf, L. (2013). *Model uncertainty and VaR aggregation*. *Journal of Banking & Finance*, 37(8), 2750–2764.
- ▣ Engle, R.F., Manganelli, S. (2004). *CAViaR: conditional autoregressive value at risk by regression quantiles*. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- ▣ Ferro, C.A.T., Segers, J. (2003). *Inference for clusters of extreme values*. *Journal of the Royal Statistical Society: Series B*, 65(2), 545–556.
- ▣ Fisher, R.A., Tippett, L.H.C. (1928). *Limiting forms of the frequency distribution of the largest or smallest member of a sample*. *Mathematical Proceedings of the Cambridge Philosophical Society*, 24(2), 180–190.
- ▣ Fissler, T., Ziegel, J.F. (2016). *Higher order elicibility and Osband's principle*. *Annals of Statistics*, 44(4), 1680–1707.
- ▣ Francq, C., Zakoïan, J.-M. (2015). *Risk-parameter estimation in volatility models*. *Journal of Econometrics*, 184(1), 158–173.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Giacomini, R., Komunjer, I. (2005). *Evaluation and combination of conditional quantile forecasts*. *Journal of Business & Economic Statistics*, 23(4), 416–431.
- ▣ Gnedenko, B. (1943). *Sur la distribution limite du terme maximum d'une série aléatoire*. *Annals of Mathematics*, 44(3), 423–453.

References (4/6)

- ▣ Gneiting, T. (2011). [Making and evaluating point forecasts](#). *Journal of the American Statistical Association*, 106(494), 746–762.
- ▣ Gumbel, E.J. (1958). [Statistics of Extremes](#). Columbia University Press.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). [Applied Quantitative Finance](#), 3rd ed. Springer.
- ▣ Harvey, D., Leybourne, S., Newbold, P. (1997). [Testing the equality of prediction mean squared errors](#). *International Journal of Forecasting*, 13(2), 281–291.
- ▣ Hill, B.M. (1975). [A simple general approach to inference about the tail of a distribution](#). *Annals of Statistics*, 3(5), 1163–1174.
- ▣ Hull, J., White, A. (1998). [Incorporating volatility updating into the historical simulation method for value-at-risk](#). *Journal of Risk*, 1(1), 5–19.
- ▣ J.P. Morgan/Reuters (1996). [RiskMetrics – Technical Document](#), 4th ed. New York.
- ▣ Koenker, R., Bassett, G. (1978). [Regression quantiles](#). *Econometrica*, 46(1), 33–50.
- ▣ Longin, F.M. (2000). [From value at risk to stress testing: the extreme value approach](#). *Journal of Banking & Finance*, 24(7), 1097–1130.
- ▣ McNeil, A.J., Frey, R. (2000). [Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach](#). *Journal of Empirical Finance*, 7(3–4), 271–300.

References (5/6)

- ▣ McNeil, A.J., Frey, R., Embrechts, P. (2015). *Quantitative Risk Management: Concepts, Techniques and Tools*, revised ed. Princeton University Press.
- ▣ Newey, W.K., West, K.D. (1987). *A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix.* *Econometrica*, 55(3), 703–708.
- ▣ Newey, W.K., West, K.D. (1994). *Automatic lag selection in covariance matrix estimation.* *Review of Economic Studies*, 61(4), 631–653.
- ▣ Nolde, N., Ziegel, J.F. (2017). *Elicitability and backtesting: perspectives for banking regulation.* *Annals of Applied Statistics*, 11(4), 1833–1874.
- ▣ Patton, A.J., Ziegel, J.F., Chen, R. (2019). *Dynamic semiparametric models for expected shortfall (and Value-at-Risk).* *Journal of Econometrics*, 211(2), 388–413.
- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). *Information entropy and measures of market risk.* *Entropy*, 19(5), 226.
- ▣ Pele, D.T., Lazar, E., Mazurencu-Marinescu-Pele, M. (2019). *Modeling expected shortfall using tail entropy.* *Entropy*, 21(12), 1204.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). *Using high-frequency entropy to forecast Bitcoin's daily value at risk.* *Entropy*, 21(2), 102.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). *Finite-sample precision limits for expected shortfall forecast comparisons.* *Mathematics*, 14(13), 2316.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). *Are cryptos becoming alternative assets?.* *European Journal of Finance*, 29(10), 1064–1105.

References (6/6)

- ▣ Pickands, J. (1975). Statistical inference using extreme order statistics. *Annals of Statistics*, 3(1), 119–131.
- ▣ Rockafellar, R.T., Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2(3), 21–41.
- ▣ Rockafellar, R.T., Uryasev, S. (2002). Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance*, 26(7), 1443–1471.
- ▣ Spierdijk, L. (2016). Confidence intervals for ARMA–GARCH value-at-risk: the case of heavy tails and skewness. *Computational Statistics & Data Analysis*, 100, 545–559.
- ▣ Tasche, D. (2008). Capital allocation to business units and sub-portfolios: the Euler principle. arXiv:0708.2542.
- ▣ Taylor, J.W. (2019). Forecasting value at risk and expected shortfall using a semiparametric approach based on the asymmetric Laplace distribution. *Journal of Business & Economic Statistics*, 37(1), 121–133.
- ▣ Yamai, Y., Yoshida, T. (2005). Value-at-risk versus expected shortfall: a practical perspective. *Journal of Banking & Finance*, 29(4), 997–1015.